Graphical Abstract

Regression-Based Explainable Deep Learning for Estimating Hamiltonian Parameters from Magnetic Nanodot Images

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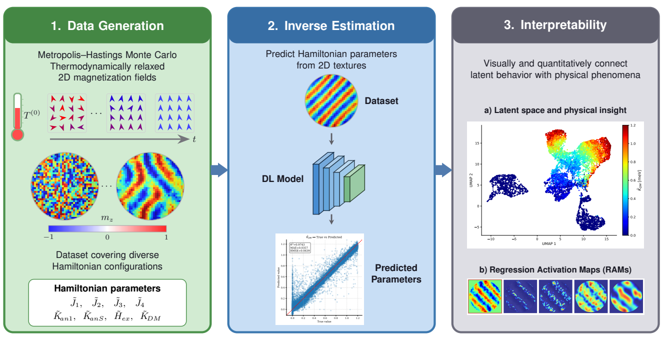

Highlights

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- Atomistic Hamiltonian generates relaxed 2D magnetic domain images for training.

- Deep learning regression recovers Hamiltonian parameters from nanodot images.

- Regression Activation Maps explain continuous prediction errors spatially.

- Framework isolates degenerate textures linking black-box AI to magnetic physics.

Regression-Based Explainable Deep Learning for Estimating Hamiltonian Parameters from Magnetic Nanodot Images

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Abstract

In magnetic nanodots, the stability of emergent topological textures is governed by an energetic competition described by the extended Heisenberg Hamiltonian, involving symmetric exchange, the Dzyaloshinskii–Moriya interaction (DMI), magnetocrystalline anisotropy, and the Zeeman field. Recovering these atomistic parameters from two-dimensional magnetic domain images is a severely ill-posed inverse problem, since distinct parameter combinations yield states with nearly indistinguishable free energies. Empirical fitting and simulation-based searches struggle to discriminate among competing solutions, while standard deep learning regressors collapse the multimodal inverse landscape into opaque, oversmoothed estimates. We propose an inherently explainable regression framework that couples Metropolis–Hastings Monte Carlo sampling with simulated annealing, used to generate thermodynamically relaxed magnetization fields, to convolutional and Vision Transformer architectures trained to recover eight Hamiltonian parameters. Interpretability is delivered by Regression Activation Maps (RAMs), a novel attribution method extending gradient-based saliency to continuous physical targets. On 218,256 simulated configurations, the framework predicts the annealing temperature, second-shell exchange constant, and DMI strength with R^2 of 0.91, 0.93, and 0.97. Unsupervised clustering of the latent space recovers four magnetic phases without supervision, and RAMs isolate chiral domain walls and skyrmion cores as the textures driving predictions, establishing a traceable link between atomistic design parameters and observable spin textures.

Keywords: Explainable AI, Magnetic Nanodots, Hamiltonian Parameters, Regression Activation Maps, Deep Learning

1. Introduction

Understanding and controlling magnetic behavior at the nanometric scale is a central challenge in modern condensed matter physics and materials engineering [1]. This capability is particularly relevant to the development of spintronic devices, ultrahigh-density magnetic memories, and neuromorphic computing platforms based on topological magnetic textures [2, 3]. At such length scales, surface effects, geometric confinement, and energy quantization substantially modify magnetic properties relative to the bulk [4]. The resulting enhancement of surface anisotropy, structural imperfections, and thermal fluctuations strongly affects the stability and dynamics of spin configurations [5]. These effects give rise to emergent collective states—including ferromagnetic, helical, labyrinthine, and skyrmionic textures [6, 7, 8, 9]—whose stability is governed by the interplay of competing microscopic interactions [10]. In this context, magnetic nanodots constitute an ideal model platform, as their lateral confinement favors the formation of single-domain states, vortices, and skyrmion-like configurations controlled by the balance among symmetric exchange, magnetic anisotropy, DMI, and external magnetic fields [11, 12]. Therefore, inferring these underlying physical parameters directly from magnetic domain images has become a critical inverse problem for the rational engineering of functional magnetic materials [13].

Within this framework, atomistic Hamiltonian-based modeling constitutes the most rigorous description of the discrete spin interactions, localized dynamics, and boundary effects that dominate nanoscale magnetic systems. Unlike continuum micromagnetic models, which can mask essential surface and finite-size contributions, the extended Heisenberg Hamiltonian explicitly captures the atomistic interactions responsible for stabilizing complex topological textures in magnetic nanodots [14]. In its general form, the Hamiltonian includes several competing terms:

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the symmetric exchange interaction ($\tilde{J}$ [meV]), which promotes collinear alignment; the DMI ($\tilde{D}$ [meV]), which favors chiral and non-collinear spin textures; the magnetocrystalline anisotropy ($\tilde{K}$ [meV]), which selects preferred crystallographic orientations; and the Zeeman coupling to an external magnetic field ($\tilde{H}$ [meV]) [15]. The competition among these contributions generates a highly multimodal energy landscape [10, 16], in which physically distinct parameter sets may produce spin states with nearly indistinguishable free energies. This degeneracy can lead to the coexistence of helical, skyrmion-like, and other metastable textures [17, 18], thereby rendering the inverse estimation of Hamiltonian parameters from two-dimensional magnetic domain images intrinsically ill-conditioned and structurally ambiguous [19, 20].

The intrinsic ambiguity of the inverse problem is further exacerbated by the limitations of current experimental and analytical methodologies. Although advanced imaging techniques, including magnetic force microscopy, Lorentz transmission electron microscopy, and spin-polarized scanning tunneling microscopy [21], enable high-resolution visualization of magnetic textures, they typically provide indirect or transformed measurements that are subject to modality-dependent distortions, noise, and finite spatial resolution [22, 23]. In practice, parameter estimation therefore relies heavily on empirical fitting or on comparison with synthetic magnetic configurations generated by atomistic spin simulations or continuum micromagnetic models [24, 25]. Yet similarity measures defined in image space—such as mean squared error or structural similarity—are often inadequate for resolving topologically distinct states, since they primarily quantify visual resemblance rather than physical equivalence. Consequently, the optimization process remains structurally ambiguous, and substantially different Hamiltonian parameter sets may produce magnetic domain patterns that are visually indistinguishable [26, 27].

To circumvent the prohibitive cost of iterative simulation-based optimization, recent efforts have reformulated this inverse problem as a data-driven learning task using deep neural architectures [28]. Although supervised regression models allow fast parameter inference, they generally impose an implicit one-to-one mapping and therefore tend to collapse the intrinsic multimodality of degenerate magnetic systems into oversmoothed point predictions [29]. Methods designed to explicitly capture this uncertainty—such as Bayesian inference, diffusion models, cycle-consistent networks, and physics-informed neural networks—offer improved representational flexibility, but often introduce substantial computational overhead and pronounced optimization stiffness [30, 31, 32]. Beyond these computational challenges, a central limitation remains their opacity. In strongly degenerate parameter spaces, conventional black-box models offer little or no mechanistic understanding of how localized spatial features contribute to the inferred Hamiltonian parameters [33, 34]. Moreover, although gradient-based attribution tools such as class activation maps are widely used in classification, analogous interpretability methods for continuous regression in highly multimodal physical systems remain underdeveloped [35]. This interpretability gap restricts the physical traceability required for robust and scientifically meaningful machine-learning-based inference.

Recent advances have demonstrated the immense potential of deep learning in resolving the inverse problem of magnetic Hamiltonian parameter estimation from domain images. For example, Kwon et al. [20] employed ResNet architectures to estimate Dzyaloshinskii-Moriya, anisotropy, and dipolar interactions from Monte Carlo-simulated spin configurations, while Kong et al. [13] utilized customized convolutional neural networks to extract exchange stiffness, effective anisotropy, and interfacial DMI from micromagnetic simulations. More recently, Lee et al. [28] demonstrated the applicability of fully connected networks for estimating interlayer and intralayer interactions in twisted van der Waals magnets. Despite their impressive predictive accuracy, these pioneering approaches predominantly treat the neural network as a "black box." Specifically, while some works analyze intermediate feature maps [13] or perform statistical error analysis [20, 28], they lack spatially resolved attribution mechanisms to explicitly identify which morphological features of the spin textures drive specific parameter predictions. Furthermore, physical identifiability limits—such as degenerate spin configurations arising from competing symmetric interactions—are often encountered as empirical estimation errors rather than explicitly quantified boundaries. In this work, we address these critical gaps by integrating modern deep learning architectures (DenseNet121 and Vision Transformers) with a comprehensive Explainable AI (XAI) framework. By employing Regression Activation Mapping (RAM) and UMAP/HDBSCAN clustering, we provide direct spatial mappings of the network's attention on specific magnetic textures (e.g., skyrmions, domain walls). Moreover, we explicitly establish the physical identifiability boundaries for an 8-parameter extended Heisenberg Hamiltonian, thereby bridging the gap between high-throughput parameter regression and interpretable microscopic physics.

To address these limitations, we propose an inherently explainable framework grounded in an atomistic extended Heisenberg Hamiltonian. The proposed methodology explicitly captures the competition among symmetric exchange, DMI, magnetocrystalline anisotropy, and Zeeman coupling through Metropolis–Hastings Monte Carlo sampling and simulated annealing, enabling the generation of thermodynamically relaxed and highly accurate two-dimensional magnetization fields. These physically grounded data are subsequently integrated with a regression-based deep learning architecture for robust inverse estimation of the governing Hamiltonian parameters. The main contributions of this work are summarized as follows. First, we develop an atomistic Hamiltonian simulation framework based on the extended Heisenberg

Hamiltonian, in which stochastic Monte Carlo sampling and simulated annealing are used to generate thermodynamically relaxed two-dimensional magnetization images and to establish an explicit correspondence between physical parameters and their associated topological magnetic states. Second, we integrate advanced convolutional neural networks and vision transformer architectures into a generalized regression framework for the robust prediction of fundamental Hamiltonian parameters from magnetic domain images. Third, we introduce RAMs, a novel interpretability framework that extends gradient-based activation mapping to continuous-valued physical targets, thereby providing both visual and quantitative insight into the relationship between the network's latent representations and the underlying magnetic phenomena. Finally, we demonstrate that RAMs can isolate degenerate magnetic textures, such as chiral domain walls and skyrmions, that most strongly govern the prediction of specific Hamiltonian parameters, thus providing a physically traceable alternative to conventional deterministic models whose predictive mechanisms remain opaque.

To assess the effectiveness of the proposed framework, extensive experiments were performed using a custom-generated dataset of simulated magnetic domain images covering a broad range of Hamiltonian parameter configurations. The obtained results indicate that the proposed method enables accurate inverse prediction of key magnetic parameters, with the DenseNet121 architecture achieving the strongest overall performance among the tested models. In particular, the framework shows high predictive capability for the DMI and symmetric exchange parameters. Moreover, the qualitative and quantitative analyses based on RAMs indicate that the learned representations are associated with physically relevant spin topologies rather than dataset-specific noise or superficial correlations, thereby strengthening the interpretability and reliability of the proposed approach. The remainder of the paper is organized as follows. Section 2 presents the related work. Section 3 details the materials and methods. Section 4 describes the experimental setup. Section 5 presents and discusses the results. Finally, Section 6 concludes the paper and highlights future research directions.

2. Related Work

The inverse estimation of Hamiltonian parameters from magnetic domain configurations is widely recognized as a highly challenging problem due to the complex, nonlinear relationship between physical interactions and observable spin textures. From an experimental standpoint, researchers heavily rely on advanced indirect imaging techniques such as magnetic force microscopy, Lorentz transmission electron microscopy, and spin-polarized scanning tunneling microscopy [21]. While these modalities enable high-resolution visualization of magnetic domains, they inherently capture transformed or partial projections of the underlying three-dimensional spin configurations. Because each technique is governed by distinct physical interaction principles, experimental observations are frequently contaminated by modality-specific distortions, finite spatial resolution limits, and heterogeneous noise, making direct inversion from these measurements generally infeasible [22, 23].

Consequently, practical parameter estimation heavily depends on empirical fitting or direct comparison with synthetic data generated through numerical simulations [36, 37]. Computationally, the problem is most commonly addressed using physics-based forward simulations, specifically atomistic spin models governed by the extended Heisenberg Hamiltonian and continuum micromagnetic models [24, 25, 38]. However, attempting to optimize the inverse mapping by minimizing pixel-level discrepancy functions—such as Mean Squared Error or Mean Absolute Error—frequently fails to capture physically meaningful topological differences or vital spin interactions [39, 40]. Even structural metrics like the Structural Similarity Index fall short in this regard [27]. Ultimately, because the comparison between simulated and experimental data occurs strictly in the image space rather than the parameter space, the optimization landscape becomes highly ambiguous, allowing disparate parameter configurations to yield visually identical domain patterns [26].

To bypass the computationally prohibitive nature of iterative simulation-based optimization, recent advances have increasingly framed the inverse problem as a data-driven mapping utilizing deep learning architectures [28]. The most prevalent approach employs supervised regression models, including convolutional neural networks and transformer-based architectures, to approximate this mapping deterministically. While these models offer rapid inference and scalability, they inherently assume a one-to-one mapping between images and parameters. In highly degenerate magnetic systems, this assumption causes the models to collapse the intrinsic multimodality of the parameter space into a single, often smoothed, point estimate, rendering them highly sensitive to dataset bias and sim-to-real mismatch [29].

To explicitly model uncertainty and capture this multimodality, probabilistic frameworks such as Bayesian inference methods, conditional variational autoencoders, and diffusion models have been introduced [30, 41, 42, 43]. Despite their theoretical flexibility in sampling multiple plausible configurations, these generative and Bayesian approaches suffer from immense computational overhead, unstable training dynamics, and a distinct lack of explicit physical guarantees. In parallel, cycle-consistent frameworks and Physics-Informed Neural Networks attempt to constrain the solution space by enforcing coherence between forward and inverse mappings, or by directly embedding thermodynamic penalties into the loss function [31, 44, 32]. While physically plausible, these hybrid models introduce severe optimization stiffness and rely fundamentally on the accuracy of the underlying forward surrogate, which can dangerously propagate errors within strongly degenerate spin regimes.

Despite the growing predictive capabilities of advanced deep neural networks, their fundamental opacity remains a critical barrier to their widespread adoption in condensed matter physics. In the context of nanoscale magnetism, the delicate energetic competition among the exchange interaction, the DMI, and magnetocrystalline anisotropy creates a densely degenerate parameter space where multiple physical configurations present nearly identical free energies [34, 45]. Traditional black-box deep learning architectures provide no mechanistic insight into how specific spatial features within a magnetic domain image correlate with the predicted Hamiltonian constants [33, 46]. This lack of interpretability prevents researchers from distinguishing whether a model has genuinely captured the underlying physical laws or merely memorized superficial dataset artifacts. While gradient-based attribution methods like Class Activation Maps have been widely adopted for discrete classification tasks, interpreting continuous regression outputs in highly multimodal physical systems remains largely underexplored [35]. The inability to spatially trace and quantify the sources of prediction errors restricts the scientific traceability of these models, underscoring an urgent necessity for architectures that inherently couple high predictive accuracy with transparent physics-informed spatial interpretations.

To overcome the limitations, our methodology is fundamentally grounded in a rigorous atomistic extended Heisenberg Hamiltonian framework. By explicitly modeling the delicate energetic competition among symmetric exchange, DMI, magnetocrystalline anisotropy, and Zeeman coupling through Metropolis-Hastings Monte Carlo sampling and simulated annealing, we computationally stabilize and extract highly accurate, thermodynamically relaxed two-dimensional spatial magnetization fields. In contrast to prevalent black-box methodologies and computationally prohibitive probabilistic frameworks, we directly couple these physically exhaustive simulations with an inherently explainable, regression-based deep learning architecture explicitly designed for the rigorous inverse estimation of these governing parameters. By integrating advanced Convolutional and Visual Transformer networks, we robustly tackle the structural ambiguity of the inverse problem without sacrificing architectural transparency. Also, we introduce a RAMs, an interpretability approach that directly correlates localized spatial activations within the simulated magnetic domain visualizations to the absolute prediction error of the continuous physical variables. Unlike standard structural metrics or deterministic regression models that obscure the origin of their predictions, our method visually and quantitatively bridges the latent behavior of the neural network with the rigorously modeled phenomena. This enables the precise isolation of degenerate magnetic textures—such as chiral domain walls and skyrmions—that drive specific parameter estimations, establishing a highly reliable, transparent and traceable pathway for analyzing complex nanoscale magnetic systems.

3. Materials and Methods

3.1. Atomistic Hamiltonian Framework

The theoretical foundation for modeling interacting magnetic dipole moments originates from the seminal formulations of Ernst Ising and Werner Heisenberg [47]. Ising introduced a one-dimensional statistical mechanics model wherein localized spins were heavily constrained to discrete, scalar states (i.e., rigidly pointing "up" or "down"). While the Ising model provided early fundamental insights into macroscopic ferromagnetism and phase transitions, its strict uni-axial constraint renders it structurally insufficient for capturing the continuous rotational degrees of freedom required to model complex, multidimensional magnetic textures. This limitation was subsequently resolved by Heisenberg, who formulated a generalized model allowing spin vectors to vary continuously in three-dimensional space with rotational symmetry. While the foundational Heisenberg model successfully captures isotropic exchange driven by the Pauli exclusion principle, modern nanoscale applications—such as the study of magnetic vortices and topologically protected skyrmions in confined nanodot geometries—necessitate an extended approach [3]. This extended framework rigorously incorporates symmetry-breaking relativistic spin-orbit coupling effects, primarily the DMI and magnetocrystalline anisotropy, alongside standard Zeeman coupling [48].

Formally, let $\mathcal{V} = \{1, \ldots, N\}$ denote the set of indices for the discrete magnetic lattice sites. The thermodynamic state of the system is characterized by the spatial distribution of localized magnetic moments, represented by the normalized spin vectors $\mathbf{s}_i \in \mathbb{R}^3$, subject to the constraint $\|\mathbf{s}_i\|_2 = 1$, for $i \in \mathcal{V}$. In a generalized theoretical context, all pairwise symmetric and antisymmetric interactions, as well as site-specific field and anisotropy alignments, can be collapsed into a unified geometric approach [49, 50]. The total effective energy of the system $\mathcal{H} \in \mathbb{R}$ [meV] is thus defined by the following generalized quadratic form:

$$\mathcal{H}\left(\mathbf{S}|\breve{\theta}\right) = -\sum_{i=1}^{N}\sum_{j\in\mathcal{N}(i)}\langle \mathbf{s}_i, \tilde{\mathbf{M}}_{ij}\mathbf{s}_j\rangle - \sum_{i=1}^{N}\langle \tilde{\mathbf{h}}_i, \mathbf{s}_i\rangle - \sum_{i=1}^{N}\langle \mathbf{s}_i\tilde{\mathbf{K}}_i, \mathbf{s}_i\rangle, \quad (1)$$

where $\mathcal{N}(i)$ defines the set of interacting nearest neighbors for a given site i, and the matrix $\mathbf{S} \in \mathbb{R}^{N\times 3}$ gathers the continuous spin vectors across the lattice with $\mathbf{s}_i \in \mathbf{S}$. Here, $\breve{\theta} = \{\tilde{\mathbf{M}}_{ij}, \tilde{\mathbf{h}}_i, \tilde{\mathbf{K}}_i\}$, where $\tilde{\mathbf{M}}_{ij} \in \mathbb{R}^{3\times 3}$ represents the complete bilinear interaction matrix coupling neighboring spins, $\tilde{\mathbf{h}}_i \in \mathbb{R}^3$ accounts for linear external field interactions, and $\tilde{\mathbf{K}}_i \in \mathbb{R}^{3\times 3}$ is the anisotropy tensor governing site-specific directional preferences.

Afterward, to recover the isolated physical phenomena and define an extended target scalar parameter space $\boldsymbol{\theta} = [\tilde{J}_1, \tilde{J}_2, \tilde{J}_3, \tilde{J}_4, \tilde{D}, \tilde{H}, \tilde{K}_{an1}, \tilde{K}_{anS}]$, the generalized matrices must be mathematically decomposed. By factorizing these parameters, we can map each matrix operation to its specific energetic contribution:

– Bilinear Coupling ($\tilde{\mathbf{M}}_{ij}$): A real square matrix can be separated into its symmetric and skew-symmetric components. Here, $\tilde{\mathbf{M}}_{ij}$ is parameterized by factorizing the isotropic exchange scalar $\tilde{J}_r$ for the r-th neighbor shell and the DMI magnitude $\tilde{D}$ along a predefined spatial direction vector $\hat{\mathbf{d}}_{ij} \in \mathbb{R}^3$ ($\|\hat{\mathbf{d}}_{ij}\|_2 = 1$), yielding:

$$\tilde{\mathbf{M}}_{ij}(\tilde{J}_r, \tilde{D}) = \tilde{J}_r \mathbf{I} - \tilde{D}[\hat{\mathbf{d}}_{ij}]_\times, \tag{2}$$

where $\mathbf{I} \in \mathbb{R}^{3\times3}$ represents the identity matrix and $[\hat{\mathbf{d}}_{ij}]_\times \in \mathbb{R}^{3\times3}$ denotes the skew-symmetric matrix representation of the cross-product. The symmetric exchange is modeled across four neighbor shells $r \in \{1, 2, 3, 4\}$, where $\mathcal{N}_r(i)$ denotes the set of neighbors at shell r around site i, ordered by increasing interatomic distance ($d_1 < d_2 < d_3 < d_4$). To resolve the fundamental scale ambiguity of the inverse problem—whereby multiplying all parameters by a common factor leaves the observable spin configuration unchanged—the first-shell exchange is fixed as the energy reference unit, $\tilde{J}_1 \in \mathbb{R}$ in meV, while $\tilde{J}_2, \tilde{J}_3, \tilde{J}_4 \in \mathbb{R}$ are free parameters (also in meV). The DMI acts exclusively between first-shell neighbors and is therefore indexed only over $\mathcal{N}_1(i)$.

– Zeeman Interaction ($\tilde{\mathbf{h}}_i$): The linear field term is factorized by extracting the scalar field magnitude $\tilde{H}$ and the fixed normalized direction vector $\hat{\mathbf{h}} = \hat{\mathbf{e}}_x$, with $\|\hat{\mathbf{e}}_x\|_2 = 1$, of the externally applied magnetic field, yielding $\tilde{\mathbf{h}}_i = \tilde{H}\,\hat{\mathbf{e}}_x \in \mathbb{R}^3$. Fixing the field direction along $\hat{\mathbf{e}}_x$ eliminates the orientational degree of freedom of $\tilde{\mathbf{h}}_i$, ensuring that $\tilde{H}$ alone serves as the free parameter governing the Zeeman contribution.

– Magnetocrystalline Anisotropy ($\tilde{\mathbf{K}}_i$): Modeling cubic magnetocrystalline anisotropy requires factorizing a site-dependent scalar constant $\tilde{K}_i^{eff}$ to scale a quartic interaction. The principal crystallographic axes of the nanodot are defined by the orthonormal basis vectors $\mathbf{e}_{\hat{x}}, \mathbf{e}_{\hat{y}}, \mathbf{e}_{\hat{z}} \in \mathbf{E}$, with $\mathbf{E} \in \mathbb{R}^{3\times3}$, and the directional cosines are given by $s_{im} = \langle \mathbf{e}_m, \mathbf{s}_i \rangle$ ($\forall m \in \{\hat{x}, \hat{y}, \hat{z}\}$). Because atoms at the nanodot surface experience broken inversion symmetry—having fewer neighbors on the exterior side—their anisotropy differs fundamentally from that of interior atoms. The lattice is therefore partitioned into two disjoint subsets: the bulk region $\mathcal{V}_{bulk} \subset \mathcal{V}$, comprising interior sites with complete coordination shells, and the surface region $\mathcal{V}_{surf} \subset \mathcal{V}$, comprising boundary sites exposed to the geometric confinement, with $\mathcal{V}_{bulk} \cup \mathcal{V}_{surf} = \mathcal{V}$ and $\mathcal{V}_{bulk} \cap \mathcal{V}_{surf} = \emptyset$. This yields the site-dependent effective anisotropy constant:

$$\tilde{K}_i^{eff} = \begin{cases} \tilde{K}_{an1} & \text{if } i \in \mathcal{V}_{bulk}, \\ \tilde{K}_{anS} & \text{if } i \in \mathcal{V}_{surf}, \end{cases} \tag{3}$$

where $\tilde{K}_{an1}, \tilde{K}_{anS} \in \mathbb{R}$ are scalar constants measured in meV per atom [51].

Substituting these factorized relationships back into Equation 1 yields the following explicit formulation for the total extended Hamiltonian:

$$\mathcal{H}\left(\mathbf{S}|\hat{\mathbf{d}}_{ij}, \hat{\mathbf{e}}_x; \boldsymbol{\theta}\right) = -\sum_{r=1}^{4} \tilde{J}_r \sum_{i=1}^{N} \sum_{j\in\mathcal{N}_r(i)} \langle \mathbf{s}_i, \mathbf{s}_j \rangle + \tilde{D} \sum_{i=1}^{N} \sum_{j\in\mathcal{N}_1(i)} \langle \hat{\mathbf{d}}_{ij}, (\mathbf{s}_i \times \mathbf{s}_j) \rangle$$
$$- \tilde{H} \sum_{i=1}^{N} \langle \hat{\mathbf{e}}_x, \mathbf{s}_i \rangle + \frac{1}{S_o^4} \sum_{i=1}^{N} \tilde{K}_i^{eff} \left(s_{i\hat{x}}^2 s_{i\hat{y}}^2 + s_{i\hat{x}}^2 s_{i\hat{z}}^2 + s_{i\hat{y}}^2 s_{i\hat{z}}^2 \right),$$

with $\tilde{J}_1 = 1.0$ meV fixed as the energy reference unit. The cubic anisotropy form in the last term follows directly from expanding $\frac{1}{2}(1 - \|\mathbf{s}_i\mathbf{E}\|_4^4)$ under the L_2 normalization constraint $\|\mathbf{s}_i\|_2 = 1$, which yields $s_{i\hat{x}}^2 s_{i\hat{y}}^2 + s_{i\hat{y}}^2 s_{i\hat{z}}^2 + s_{i\hat{z}}^2 s_{i\hat{x}}^2$, normalized by S_o^4 to maintain dimensional consistency across spin magnitudes.

Physically, the extended parameter space $\boldsymbol{\theta}$ actively governs the competing energetic contributions that shape the magnetic energy landscape. Each $\tilde{J}_r$ dictates the collinear alignment tendency between spin pairs separated by the corresponding distance [52]: positive values ($\tilde{J}_r > 0$) favor parallel, ferromagnetic alignment, while negative values ($\tilde{J}_r < 0$) drive antiparallel, antiferromagnetic order. This multi-shell formulation is essential in confined nanodot geometries, where longer-range exchange pathways–mediated through intermediate atomic orbitals–contribute non-negligibly to the global energetic competition and can selectively stabilize or destabilize complex topological textures beyond what first-shell interactions alone would predict [50]. Among these constants, $\tilde{J}_1$ plays a privileged role: it corresponds to the strongest and most spatially direct exchange pathway, coupling each spin to its nearest neighbors in the cubic lattice and establishing the dominant energy scale of the magnetic system. For this reason, $\tilde{J}_1$ is fixed to 1.0 meV throughout all simulations, serving as the absolute energy reference unit against which all remaining interaction strengths are expressed. The remaining constants $\tilde{J}_2$, $\tilde{J}_3$, and $\tilde{J}_4$ are free parameters whose magnitudes decay with increasing shell distance, consistent with the exponential attenuation of quantum mechanical exchange integrals with interatomic separation [52].

The DMI magnitude, represented by $\tilde{D}$, is responsible for driving chiral, non-collinear spin winding in systems exhibiting broken spatial inversion symmetry [53]. By actively competing with the dominant first-shell exchange $\tilde{J}_1$, this antisymmetric contribution stabilizes complex topological textures, such as chiral domain walls and skyrmions. It acts exclusively between first-shell neighbors, as longer-range antisymmetric exchange pathways are negligibly small compared to the dominant spin-orbit coupling at nearest-neighbor interfaces. Measured in meV per interacting spin pair, its magnitude is fundamentally dependent on the strength of the spin-orbit coupling at the material's interfaces, with typical values spanning $[-0.5, 0.5]$ meV as a fraction of the first-shell exchange energy [53].

The Zeeman coupling energy, parameterized as $\tilde{H}$, quantifies the interaction between the localized atomic spins and the externally applied magnetic field [14]. It systematically penalizes magnetic misalignment with the fixed crystallographic axis $\hat{\mathbf{e}}_x$, frequently serving as the primary thermodynamic driving force for macroscopic magnetic phase transitions. This parameter is expressed in meV per atom, mathematically representing the effective energy equivalence ($g\mu_B B$) that transitions a macroscopic magnetic field B measured in Tesla into the atomistic energy regime. Its boundaries range from 0.0 meV to approximately 2.0 meV, directly correlating to laboratory-scale magnetic fields between 0 and 10 Tesla. Also, fixing the field direction along $\hat{\mathbf{e}}_x$ reduces the dimensionality of the inverse problem by eliminating the orientational degree of freedom of $\tilde{\mathbf{h}}_i$, ensuring that $\tilde{H}$ alone governs the Zeeman contribution as a single identifiable scalar.

Finally, the magnetocrystalline anisotropy encodes the intrinsic directional preference for magnetization relative to the underlying crystal lattice [51]. Arising from relativistic spin-orbit coupling, this term rigorously establishes the energetically favorable (easy) and unfavorable (hard) crystallographic axes for spin orientation within the nanodot. Both constants are measured in meV per atom and represent a relatively subtle symmetry-breaking energy compared to the dominant isotropic exchange. The bulk anisotropy constant $\tilde{K}_{an1}$ governs interior lattice sites $i \in \mathcal{V}_{bulk}$, where each atom is fully coordinated by neighbors in all spatial directions, yielding a locally symmetric energetic environment. Its conventional boundaries range from roughly 0.001 meV in soft bulk magnets to 1.0 meV in highly anisotropic hard nanomaterials [51]. The surface anisotropy constant $\tilde{K}_{anS}$, by contrast, governs boundary sites $i \in \mathcal{V}_{surf}$, where broken spatial inversion symmetry—arising from the absence of neighbors on the exterior side of the nanodot—produces an asymmetric local environment that intensifies the directional preference for spin orientation. Consequently, $|\tilde{K}_{anS}|$ is generally larger than $|\tilde{K}_{an1}|$, reflecting the enhanced symmetry-breaking at the nanodot boundary. This two-constant formulation is essential for accurately capturing the finite-size and surface-dominated effects that critically govern the stability of emergent topological textures in confined nanoscale geometries [3, 48].

3.2. Image-Based Dataset Generation From Monte Carlo Simulations

To accurately capture the physical thermodynamic behavior and algorithmically drive the system toward a low-energy, thermodynamically stable configuration of the nanodot, thermal fluctuations must be formally introduced through the framework of statistical mechanics. While the atomistic Hamiltonian $\mathcal{H}$ rigorously defines the internal magnetic energy for a given spatial configuration of spins, it evaluates the system strictly as a frozen, zero-temperature snapshot. Hence, temperature is fundamentally a macroscopic parameter that governs the statistical distribution of the discrete magnetic microstates rather than acting as a localized interaction within the Hamiltonian. Consequently, temperature T $[K]$ couples to the system via the Boltzmann distribution [50]. Formally, a magnetic microstate μ represents a specific, instantaneous realization of the continuous spin matrix, denoted as $\mathbf{S}_\mu$. The total thermodynamic energy of this specific configuration is evaluated against the static environmental vectors and physical interaction strengths, such that $\mathcal{H}(\mu) \equiv \mathcal{H}\left(\mathbf{S}_\mu | \hat{\mathbf{d}}_{ij}, \hat{\mathbf{h}}, \mathbf{E}; \boldsymbol{\theta}\right)$. At thermal equilibrium, the probability $P(\mu)$ of the system occupying this specific microstate is defined as:

$$P(\mu) = \frac{1}{\mathcal{Z}} \exp\left(-\frac{\mathcal{H}(\mu)}{k_B T}\right), \tag{4}$$

where k_B is the Boltzmann constant (expressed in meV/K). The term $\mathcal{Z}$ represents the canonical partition function, which mathematically enforces probability normalization by summing over all valid spin configurations ν (i.e., all possible state matrices $\mathbf{S}_\nu$ strictly satisfying the geometric boundary constraint $\|\mathbf{s}_i\|_2 = 1$):

$$\mathcal{Z} = \sum_{\mathbf{S}_\nu} \exp\left(-\frac{\mathcal{H}\left(\mathbf{S}_\nu | \hat{\mathbf{d}}_{ij}, \hat{\mathbf{h}}, \mathbf{E}; \boldsymbol{\theta}\right)}{k_B T}\right). \tag{5}$$

In our computational framework, this thermodynamic distribution is actively sampled using the Metropolis-Hastings Monte Carlo algorithm combined with a simulated annealing protocol [54]. This stochastic sampling ensures a rigorous exploration of the highly multimodal energy landscape characteristic of these degenerate magnetic systems. When a localized spin perturbation proposes a physical transition from the current microstate $\mu^{(t)}$ (characterized by $\mathbf{S}_{\mu^{(t)}}$) to a newly modified state $\mu^{(t+1)}$ (characterized by $\mathbf{S}_{\mu^{(t+1)}}$), the resulting change in the system's total energy is computed as the scalar difference $\Delta\mathcal{H} = \mathcal{H}(\mu^{(t+1)}) - \mathcal{H}(\mu^{(t)})$. To optimize computational efficiency, $\Delta\mathcal{H}$ is evaluated locally at the perturbed lattice site. The proposed state is subsequently accepted or rejected based on the precise transition probability:

$$P(\mu^{(t)} \to \mu^{(t+1)}) = \begin{cases} 1 & \text{if } \Delta\mathcal{H} < 0 \\ \exp\left(-\frac{\Delta\mathcal{H}}{k_B T^{(t)}}\right) & \text{if } \Delta\mathcal{H} \geq 0. \end{cases} \tag{6}$$

Physically, the instantaneous temperature parameter $T^{(t)}$ actively scales the energetic barriers within the system's vast topological energy landscape during the annealing process [54]. At elevated temperatures, the available thermal energy $k_B T^{(t)}$ frequently exceeds local energy penalties ($\Delta\mathcal{H}$), allowing the simulation to accept thermodynamically unfavorable spin alignments. This thermal noise provides the critical kinetic mechanism required to break out of metastable, localized energy minima, such as misaligned ferromagnetic domains [55]. As $T^{(t)}$ is systematically scaled down during the annealing procedure, the probability of accepting higher-energy configurations undergoes exponential decay. This controlled thermal relaxation drives the continuous spin vector field toward a low-energy, thermodynamically stable configuration consistent with the annealing schedule, promoting the stabilization of complex chiral skyrmion textures driven by the underlying DMI. We note that while simulated annealing with a finite cooling rate does not formally guarantee convergence to the absolute global minimum—particularly in DMI-driven systems where numerous metastable chiral states may coexist—the monotonic energy decrease observed across all simulated configurations (see Section 5) provides empirical evidence of thorough thermodynamic relaxation within the sampled parameter space.

Moreover, to ensure that each simulation consistently originates from a fully randomized paramagnetic phase regardless of the specific parameter configuration, the initial annealing temperature (i.e., $T^{(t)}$ at $t = 0$) is established as an exchange-normalized initial temperature. Consistent with foundational statistical mechanics [50], this starting temperature is dynamically rescaled in direct proportion to the dominant symmetric exchange energy ($\tilde{J}$). This energetic normalization guarantees that the initial thermal fluctuations are strictly sufficient to overcome the highest localized energetic barriers ($\Delta\mathcal{H}$) for that specific Hamiltonian. This effectively erases initial state bias and prevents the continuous spin vector field from prematurely collapsing into metastable local minima before the controlled cooling protocol begins.

To generate a robust and diverse dataset for deep learning evaluation, the magnetic parameters governing the Hamiltonian were systematically varied across the simulations. Key variables included the exchange interaction constant ($\tilde{J}$), the magnetocrystalline anisotropy ($\tilde{K}$), the external Zeeman field ($\tilde{H}$), and the DMI strength ($\tilde{D}$), alongside variations in the base initial temperature ($T^{(0)}$). Subsequently, the transformation of these raw, converged 3D spin states into normalized 2D image arrays followed a strict procedural pipeline:

- Geometry parsing: Spatial coordinates defining the nanodot geometry (with a fixed disk radius and thickness as Magnetic Unit Cell (MUC) measure) were established using a binary circular lattice mask.

- Spin state mapping: The topological structure was projected into two dimensions by extracting the out-of-plane s_z component of the continuous spin vectors ($\mathbf{s}_i$) across the valid lattice coordinates.

- Image rendering: The isolated configurations were rasterized onto a fixed $\check{H} \times \check{W}$ pixel grid. A divergent red–blue colormap was utilized to strictly encode opposite spin polarities, visually capturing the chiral winding from the nanodot's core to its perimeter.

- Preprocessing: Pixel intensities corresponding to the S_z magnitude were normalized to a strict interval $[0, 1]$ to ensure stable gradient flows during the deep learning optimization phase. To preserve the intrinsic physical symmetries and chiral handedness dictated by the DMI, no spatial augmentations (such as random rotations or structural flips) were applied to the dataset.

Lastly, the simulation pipeline yields a comprehensive image-based dataset, formalized as $\mathcal{D} = \{\mathbf{X}_n, \boldsymbol{\theta}_n\}_{n=1}^{\check{N}}$, which serves as the final collection of input-output pairs for the deep learning parameter estimation task. Each input tensor $\mathbf{X}_n \in [0, 1]^{\check{H}\times\check{W}}$ encodes the two-dimensional spatial magnetization field (the normalized s_z projection), while the corresponding target vector $\boldsymbol{\theta}_n \in \mathbb{R}^{\check{P}}$ stores the $\check{P}$-dimensional array of generating thermodynamic and Hamiltonian parameters (e.g., $\boldsymbol{\theta}_n = [T_n^{(0)}, \tilde{J}_{1n}, \tilde{J}_{2n}, \tilde{J}_{3n}, \tilde{J}_{4n}, \tilde{D}_n, \tilde{H}_n, \tilde{K}_{an1n}, \tilde{K}_{anSn}]$). Figure 1 depicts the main flowchart of our image-based magnetic dataset generation approach.

3.3. Deep Regression Framework for Hamiltonian Parameter Estimation

The inverse estimation of the Hamiltonian parameters and thermodynamic state from magnetic domain images is formalized as a continuous mapping problem. Given the generated dataset $\mathcal{D}$, where $\mathbf{X}_n$ represents the normalized s_z spatial magnetization field and $\boldsymbol{\theta}_n$ represents the vector of generating physical parameters, we aim to learn a parameterized non-linear mapping function $f_\Theta : \mathbb{R}^{\check{H}\times\check{W}} \to \mathbb{R}^{\check{P}}$, to find the predicted parameters $\hat{\boldsymbol{\theta}}_n \in \mathbb{R}^{\check{P}}$.

Here, f_Θ denotes a deep neural network architecture parameterized by a set of learnable weights and biases Θ. The optimal parameter set $\hat{\Theta}$ is obtained by minimizing a continuous loss function over the training distribution [56]:

$$\hat{\Theta} = \arg\min_{\Theta} \frac{1}{\check{N}} \sum_{n=1}^{\check{N}} \mathcal{L}(\boldsymbol{\theta}_n, f_\Theta(\mathbf{X}_n)), \tag{7}$$

where $\mathcal{L}$ represents the chosen regression objective, typically the Mean Squared Error (MSE) or Huber loss, which mitigates the influence of outlier topological states during optimization. By minimizing this objective, the network intrinsically learns to untangle the highly multimodal spatial features mapping to the specific energetic balances of the extended Heisenberg Hamiltonian.

To achieve robust feature extraction across varying scales of magnetic textures (e.g., from localized domain walls to global skyrmion geometries), this framework evaluates four distinct, state-of-the-art deep learning architectures: ResNet50, DenseNet, Xception, and the Visual Transformer (ViT).

- Residual Networks (ResNet50) [57]: Standard deep convolutional networks often suffer from the vanishing gradient problem as depth increases, hindering

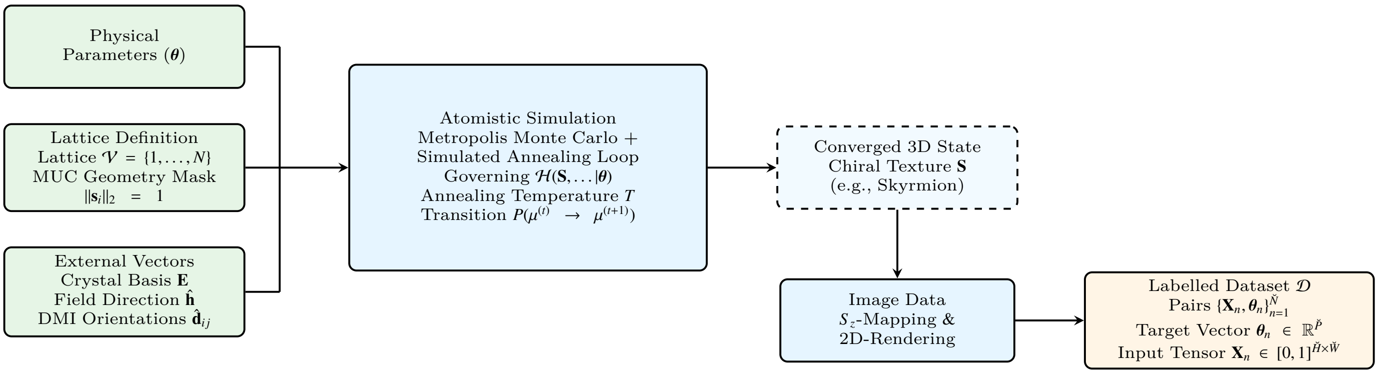

Figure 1: Main flowchart detailing the generation of the image-based magnetic dataset. The pipeline initializes with the foundational inputs (green), encompassing the Hamiltonian interaction parameters ($\boldsymbol{\theta}$), the discrete unit cell geometry, and the external directional vectors. These constraints govern the atomistic simulation (blue), which employs the Metropolis Monte Carlo algorithm coupled with simulated annealing to thermodynamically relax the system into a converged 3D topological state. Finally, the resulting continuous vector field is projected, rendered, and preprocessed (yellow) to formulate the labelled dataset $\mathcal{D}$, mapping 2D spatial magnetization images to their generating physical parameters.

the learning of complex physical mappings. ResNet50 circumvents this by introducing shortcut connections that bypass multiple convolutional layers, directly propagating spatial information and gradients. Mathematically, if $\mathbf{x}$ is the input to a sequence of layers, and $\mathcal{F}(\mathbf{x}, \{\mathbf{W}_l\})$ represents the residual mapping to be learned (comprising convolutions, batch normalization, and ReLU activations) for the l-th layer, the output $\mathbf{y}$ of the residual block is defined as:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{\mathbf{W}_l\}) + \mathbf{x}. \tag{8}$$

In the context of magnetic domain images, these skip connections allow the network to simultaneously preserve low-level spatial features (like abrupt chiral boundaries) while composing deep, high-level abstractions of the global thermodynamic state (see Figure 2).

– Densely Connected Convolutional Networks (DenseNet) [58]: While ResNet adds features via summation, DenseNet concatenates them, maximizing information flow and feature reuse throughout the network. This is particularly advantageous for the Hamiltonian inverse problem, as the delicate energetic competition often manifests in subtle, overlapping spatial frequencies that require dense multi-scale analysis. In a DenseNet block, the feature maps of all preceding layers are used as inputs to the current layer. If $\mathbf{x}_l$ denotes the output of the l-th layer, and $H_l(\cdot)$ represents a composite function of batch normalization, ReLU, and convolution, the forward pass is expressed as:

$$\mathbf{x}_l = H_l([\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_{l-1}]), \tag{9}$$

where $[\ldots]$ denotes the concatenation operation. This dense topological structure ensures that the final regression head has direct access to both fine-grained pixel interactions and coarse global structures (see Figure 3).

– Xception (Extreme Inception) [59]: This architecture operates on the hypothesis that cross-channel correlations and spatial correlations in feature maps can be entirely decoupled. It replaces standard Inception modules with depthwise separable convolutions, drastically reducing computational overhead while enhancing representational efficiency. A depthwise separable convolution splits the standard convolution into two distinct steps. First, a depthwise convolution applies a single spatial filter to each input channel independently. Second, a pointwise convolution (1×1 convolution) projects the output of the depthwise step onto a new channel space to capture cross-channel correlations. If $\mathbf{X} \in \mathbb{R}^{\tilde{H}\times\tilde{W}\times\tilde{C}}$ is the input tensor, the operation is formally approximated as:

$$\mathrm{Conv}_{\mathrm{sep}}(\mathbf{X}) = \mathrm{Conv}_{\mathrm{pointwise}}(\mathrm{Conv}_{\mathrm{depthwise}}(\mathbf{X})). \tag{10}$$

For nanoscale magnetism, this extreme decoupling allows the network to efficiently isolate specific topological frequency patterns (depthwise) before correlating them to the complex parameter space $\boldsymbol{\theta}_n$ (see Figure 4).

– Visual Transformer (ViT) [60]: Unlike Convolutional Neural Networks, which enforce a strict local inductive bias, the Vision Transformer processes the 2D spatial magnetization field $\mathbf{X}_n$ as a sequence of flattened, non-overlapping 2D patches. This allows the model to capture long-range, global dependencies across the nanodot instantly, which is critical for understanding macro-scale topological stability dictated by the Zeeman field or global anisotropy. The input image is divided into N_p patches, which are linearly

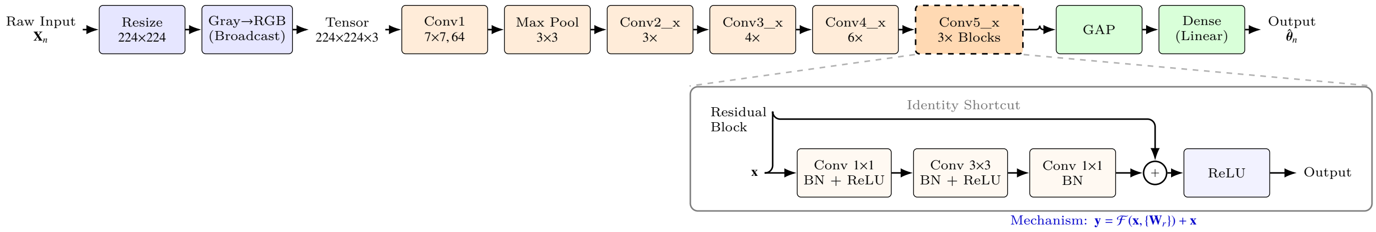

Figure 2: ResNet50 architecture pipeline for Hamiltonian parameter estimation. The pipeline progresses from single-channel input preprocessing through a pre-trained feature extractor. The Residual bottleneck structure used extensively in Conv5_x. The deepest features are processed by the customized regression head containing a Global Average Pooling (GAP) and a final dense layer with linear activation to output the predicted physical constants.

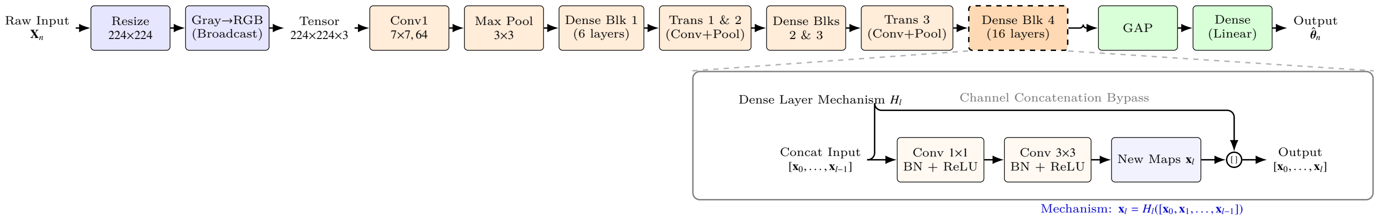

Figure 3: DenseNet121 architecture pipeline for Hamiltonian parameter estimation. The pipeline progresses left-to-right, processing single-channel inputs through the feature extractor composed of Dense Blocks and Transition layers. A detailed horizontal view illustrates the dense connectivity mechanism where feature maps from all preceding layers are concatenated before convolution. The final aggregated features are mapped to physical constants via the customized regression head.

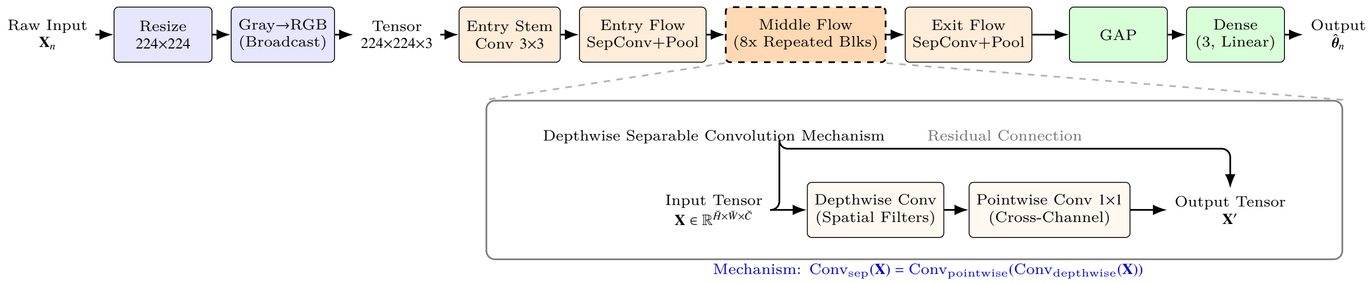

Figure 4: Xception architecture pipeline for Hamiltonian parameter estimation. The feature extractor composed of Entry, Middle, and Exit flows. A detailed horizontal view illustrates the depthwise separable convolution mechanism where spatial feature extraction (depthwise) and cross-channel correlations (pointwise) are entirely decoupled. The final aggregated features are mapped to physical constants.

projected into a sequence of D-dimensional embeddings. The core mechanism is the Multi-Head Self-Attention (MSA), which computes attention scores to weigh the importance of different patches relative to one another. For a given set of queries Q, keys K and values V derived from the patch embeddings, the scaled dot-product attention is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \qquad (11)$$

where d_k is the scaling factor (the dimension of the keys). By employing MSA, the ViT can explicitly correlate a localized structural defect on one edge of the nanodot with a compensatory chiral boundary on the opposite edge, mapping these distant correlations directly to the global DMI or symmetric exchange strengths.

To optimize the parameter set $\hat{\Theta}$ in Equation 7 across all evaluated architectures, the models are trained utilizing the MSE as the explicit continuous loss function $\mathcal{L}$. The optimization process is executed using the Adam optimizer, relying on backpropagation and automatic differentiation to efficiently compute the network gradients [19]. By iteratively updating the model weights based on these computed gradients, the training pipeline systematically minimizes the structural and energetic discrepancies between the predicted outputs and the true Hamiltonian constants, ensuring robust convergence in estimating the underlying physical parameters.

3.4. Spatial Interpretability via RAMs

Interpretability methods based on activation mapping have become a foundational approach for understanding the decision-making processes of deep neural networks. By identifying the spatial regions within an input that most significantly influence a model's prediction, these methods

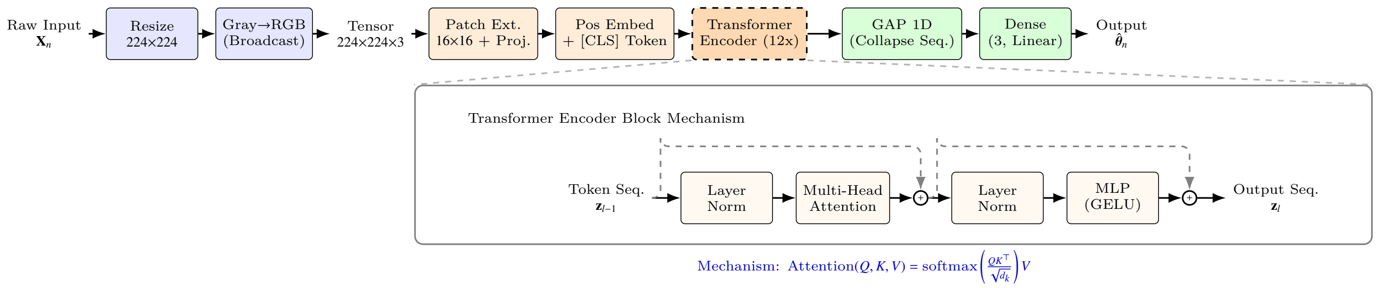

Figure 5: ViT architecture pipeline for Hamiltonian parameter estimation. The pipeline progresses left-to-right, processing single-channel inputs through patch extraction, linear projection, and a sequential backbone of Transformer Encoder blocks. The core Transformer block mechanism, highlighting the MSA that explicitly models long-range spatial dependencies across the magnetic nanodot. The final sequence of high-level token representations is aggregated via 1D GAP and mapped to the physical constants through the customized regression head.

project opaque latent representations back into a human-interpretable domain. Among these, Gradient-weighted Class Activation Mapping (Grad-CAM) and its variants have been extensively utilized in computer vision to generate class-specific saliency maps [61].

In standard classification paradigms, given an input image $\mathbf{X}$ and a discrete target class score y^c, activation maps are computed by extracting the gradients of y^c with respect to the k-th feature map $\mathbf{A}^k \in \mathbb{R}^{H' \times W'}$ of a terminal convolutional layer (where H' and W' denote the spatial dimensions of the feature map). The spatial importance weights are derived by globally pooling these gradients, and the final map is a ReLU-activated linear combination of the feature maps [62]. While highly effective for discrete semantic categorization, this standard formulation structurally fails in continuous regression tasks. The absence of a discrete target class removes the natural selection mechanism required to isolate the output dimension driving the interpretability process. Furthermore, directly computing gradients with respect to a prediction $\hat{\boldsymbol{\theta}}$ introduces severe gradient instability; minute fluctuations in the scalar output can yield drastically different spatial attributions, failing to provide the physical consistency required for analyzing complex Hamiltonian parameters [63].

3.4.1. RAMs for CNN-based Approaches

To overcome the fundamental limitations of discrete activation methods in highly degenerate physical systems, we introduce RAMs, a novel formulation explicitly tailored to bridge continuous parameter estimation with spatial topologies. Given an input magnetic domain visualization $\mathbf{X}$ and the optimized regression model $f_{\hat{\Theta}} : \mathbb{R}^{\check{H} \times \check{W}} \to \mathbb{R}^{\check{P}}$, the network predicts a vector of physical constants $\hat{\boldsymbol{\theta}} = f_{\hat{\Theta}}(\mathbf{X})$. Let $\hat{\theta}_p \in \hat{\boldsymbol{\theta}}$ denote the continuous prediction for the p-th target Hamiltonian parameter (e.g., the DMI strength $\tilde{D}$ or symmetric exchange $\tilde{J}$), and let $\theta_p \in \boldsymbol{\theta}$ represents its corresponding ground truth. Instead of utilizing a raw predicted scalar, we formulate a scale-invariant, error-based activation objective O_p. To prevent the mathematical discontinuities and negative objective bounds inherent to absolute error formulations $(1 - |\theta_p - \hat{\theta}_p|)$, we utilize a normalized Gaussian decay function:

$$O_p = \exp\left(-\gamma_p \left(\theta_p - \hat{\theta}_p\right)^2\right), \tag{12}$$

where $\gamma_p > 0$ is a hyperparameter-specific precision factor determining the strictness of the error decay. This objective smoothly bounds the activation target strictly within $(0, 1]$, promoting maximum gradient response precisely when the network's prediction asymptotically approaches the true thermodynamic value, thereby isolating the topological features that positively contribute to accurate regression.

The spatial importance weights $\alpha_k^{(p)}$ for each continuous feature map $\mathbf{A}^k$ are subsequently computed by globally averaging the gradients of the error objective O_p with respect to the local spatial activations $\mathbf{A}_{ij}^k$:

$$\alpha_k^{(p)} = \frac{1}{H'W'} \sum_{i=1}^{H'} \sum_{j=1}^{W'} \frac{\partial O_p}{\partial \mathbf{A}_{ij}^k}. \tag{13}$$

These weights strictly quantify the importance of the k-th latent spatial feature in successfully minimizing the prediction error for the target physical parameter θ_p. The resulting RAM is then formulated as:

$$\mathcal{R}_{\mathrm{RAM}}^{(p)} = \mathrm{ReLU}\left(\sum_k \alpha_k^{(p)} \mathbf{A}^k\right). \tag{14}$$

The application of the ReLU activation ensures that only spatial features exerting a positive, stabilizing influence on the estimation of the target parameter are preserved. Finally, $\mathcal{R}_{\mathrm{RAM}}^{(p)}$ is spatially upsampled to the original input resolution $\check{H} \times \check{W}$ and normalized to the interval $[0, 1]$. Then, by mapping the continuous error gradients back to the input spatial magnetization field $\mathbf{X}$, RAMs allow for the direct visual and quantitative identification of the specific emergent textures that the network utilizes to deduce the underlying energetic balances. The complete workflow for this localized spatial interpretability is illustrated in Figure 6.

3.4.2. RAMs for ViT-based Approaches

While our standard RAMs formulation seamlessly integrates with CNNs by exploiting their innate spatial hier-

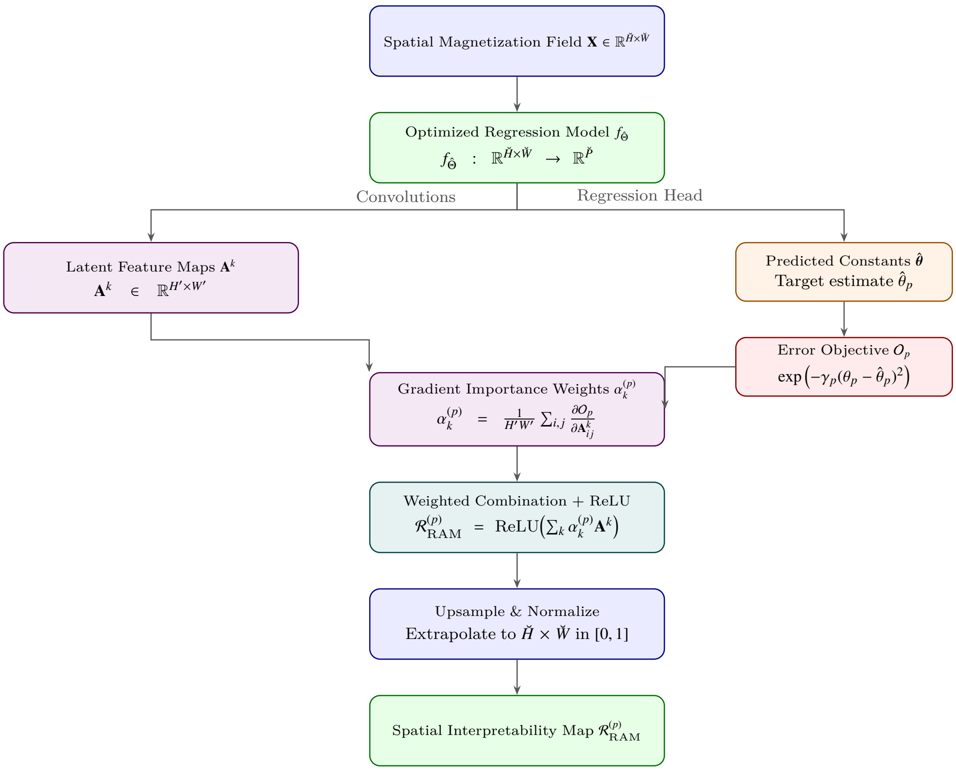

Figure 6: RAMs extraction pipeline. Latent feature maps and continuous predictions from the network backbone are explicitly coupled via an error-decay objective. Gradient-based importance weights combine these features to produce a normalized, physically traceable spatial saliency map for the target Hamiltonian parameter θ_p.

archies, extending this interpretability metric to the Vision Transformers requires mapping sequence-based latent representations back to the two-dimensional physical domain. Unlike CNNs, which preserve spatial locality through sliding convolutional filters, the ViT processes the input spatial magnetization field $\mathbf{X}$ as a flattened sequence of N_r non-overlapping patches, outputting a discrete sequence of latent embeddings. To evaluate ViT-based architectures within the exact same RAMs framework, we introduce a token-reshaping mechanism that bridges the one-dimensional transformer latent space with the two-dimensional topological geometry. Let the output of the terminal Transformer encoder block be denoted as $\mathbf{Z} \in \mathbb{R}^{(N_r+1)\times D}$, where D is the embedding dimension and the extra token corresponds to the global aggregation token (e.g., the [CLS] token) utilized by the regression head to predict the Hamiltonian parameters $\hat{\mathbf{y}}$. By isolating the N_r spatial tokens, we obtain the localized sequence $\mathbf{Z}_s \in \mathbb{R}^{N_r\times D}$. Because each sequence token explicitly corresponds to a fixed physical patch of the original nanodot domain, $\mathbf{Z}_s$ can be folded back into a 2D spatial grid. Assuming square patches, the sequence is reshaped into a pseudo-feature map:

$$\mathbf{A}_{\mathrm{ViT}} = \mathrm{Reshape}(\mathbf{Z}_s) \in \mathbb{R}^{\sqrt{N_r}\times\sqrt{N_r}\times D}. \tag{15}$$

Once projected into this tensor format, $\mathbf{A}_{\mathrm{ViT}}$ functions structurally identically to the terminal convolutional feature map $\mathbf{A}$ utilized in the CNN baseline. The k-th slice of the embedding dimension ($k \in \{1, \ldots, D\}$) forms the latent spatial map $\mathbf{A}^k_{\mathrm{ViT}} \in \mathbb{R}^{\sqrt{N_r}\times\sqrt{N_r}}$. We then apply the identical scale-invariant, error-based activation objective O_p defined in Equation 12. The spatial importance weights for the ViT embeddings are computed by globally averaging the gradients of this objective with respect to the reshaped spatial tokens:

$$\alpha_k^{(p)} = \frac{1}{N_r} \sum_{i=1}^{\sqrt{N_r}} \sum_{j=1}^{\sqrt{N_r}} \frac{\partial O_p}{\partial (\mathbf{A}_{\mathrm{ViT}})^k_{ij}}. \tag{16}$$

RAMs for the ViT are constructed via the ReLU-activated linear combination of these reshaped embedding slices, precisely mirroring Equation 14, before being upsampled to the original input resolution $\breve{H} \times \breve{W}$. This extension guarantees an architecture-agnostic evaluation of interpretability, ensuring that the isolated degenerate textures driving the model's predictions can be fairly compared across fundamentally disparate deep learning architectures.

3.4.3. Normalized RAMs for Layer-Wise and Target-Specific Analysis

To rigorously compare the spatial activations across different Hamiltonian parameters and network depths, it is imperative to compute the RAMs utilizing strictly normalized predictions and targets. Let $\bar{\theta}_p \in [0, 1]$ and $\hat{\bar{\theta}}_p \in [0, 1]$ denote the Min-Max normalized ground truth and predicted values for the p-th physical parameter, respectively. By scaling the variables to a uniform interval, the layer-specific error objective is strictly bounded and evaluated as $O_p = \exp(-\gamma_p(\bar{\theta}_p - \hat{\bar{\theta}}_p)^2)$. For a given convolutional or transformer layer $l \in \{1, \ldots, L\}$, we extract the layer-specific unnormalized RAM, denoted as $\mathcal{R}^{(l,p)}_{\mathrm{RAM}}$, following the gradient and activation protocols established in Equations 13 and 14. To quantitatively evaluate and isolate the relative contribution of each distinct layer to the estimation of every physical constant, we introduce a global cross-layer and cross-parameter normalization scheme. The globally normalized activation map, $\tilde{\mathcal{R}}^{(l,p)}_{\mathrm{RAM}} \in [0, 1]$, is defined by scaling the layer-specific maps against the absolute maximum spatial activation observed across all L layers and all $\check{P}$ parameters:

$$\tilde{\mathcal{R}}^{(l,p)}_{\mathrm{RAM}} = \frac{\mathcal{R}^{(l,p)}_{\mathrm{RAM}}}{\max\limits_{l' \in \{1,\ldots,L\}, p' \in \{1,\ldots,\check{P}\}} \left(\max\limits_{i,j} \left[\mathcal{R}^{(l',p')}_{\mathrm{RAM}} \right]_{i,j} \right)}. \quad (17)$$

This comprehensive normalization ensures that the gradient magnitudes are dimensionally consistent and comparable, preventing parameters with intrinsically higher variance or layers with larger gradient flows from disproportionately dominating the interpretability metrics. By tracking the evolution of $\tilde{\mathcal{R}}^{(l,p)}_{\mathrm{RAM}}$ across the network depth, this formulation explicitly reveals how the deep learning architecture bypasses the intrinsically ill-conditioned nature of the inverse problem. In magnetic nanodots, severe structural ambiguity arises because disparate energetic combinations of the symmetric exchange ($\tilde{J}$), DMI ($\tilde{D}$), magnetocrystalline anisotropy ($\tilde{K}$), and Zeeman field ($\tilde{H}$) can yield highly degenerate states with nearly identical free energies. Then, the global normalization aim to code how the network dynamically resolves this degeneracy through a hierarchical physical decoupling. We establish that early network layers (lower l) exhibit high relative activations for parameters governing short-range spin interactions, capturing the localized chiral winding angles and domain wall boundaries dictated primarily by the delicate competition between $\tilde{J}$ and $\tilde{D}$. Conversely, deeper layers (higher l) aggregate these localized topological textures to resolve macro-scale thermodynamic stability. These terminal layers exhibit dominant spatial activations for $\tilde{H}$ and $\tilde{K}$, which govern the global topological confinement, phase transitions, and spatial alignment of emergent structures such as skyrmion cores and labyrinthine patterns. By correlating the intensity and spatial localization of $\tilde{\mathcal{R}}^{(l,p)}_{\mathrm{RAM}}$ across the network hierarchy, our framework utilizes both localized boundary chirality and global domain spacing to disambiguate the underlying Hamiltonian parameters.

4. Experimental Set-Up

To evaluate the effectiveness of the proposed framework for Hamiltonian parameter estimation, we design a comprehensive experimental protocol that integrates physics-based simulations, data-driven modeling, and interpretability analysis. The objective of this setup is to systematically assess not only the predictive performance of the model, but also its ability to generalize across different physical regimes and provide meaningful spatial explanations through the proposed RAMs. The dataset utilized in this study consists entirely of simulated magnetic domain configurations generated through our atomistic Monte Carlo framework, as described in Section 3.2. To accurately reflect the finite-size effects and surface boundaries inherent to experimental nanostructures, the magnetic nanodots were modeled on a three-dimensional discrete lattice. The cylindrical geometry was constrained by a radius of $R_d = 18.25$ [MUC] and a thickness of $\check{L} = 5$ [MUC]. Within this confined grid, the spatial bounds were dictated by a strictly defined circular binary mask, yielding an active interacting volume of approximately 5,200 localized spins per simulated dot.

For each unique combination of Hamiltonian parameters, the system was subjected to a rigorous simulated annealing protocol to isolate the thermodynamically relaxed state. To guarantee full ergodicity, the exchange-normalized initial temperature was established at an effective paramagnetic ceiling. The system was systematically cooled to a near-ground state of $T^{(f)} = 0.1$ K utilizing a linear decrement of $\Delta T = -0.1$ K. At each discrete temperature interval, the spin lattice underwent 8,000 Monte Carlo Sweeps (MCS) exclusively dedicated to thermalization, thereby overcoming metastable localized energetic barriers. This thermalization phase was subsequently followed by 100 measurement MCS to compute macroscopic thermodynamic averages (e.g., specific heat and magnetic susceptibility) and verify strict energetic convergence. To overcome the prohibitive computational bottleneck traditionally associated with sequential Monte Carlo methods, the atomistic simulation was rigorously optimized utilizing PyTorch tensor operations on CUDA-enabled GPUs.

Upon converging to a low-energy thermodynamically stable state at 0.1 K, the final topological state was isolated. To mimic experimental 2D observation techniques, the spatial magnetization field was extracted exclusively from the central discrete layer of the nanodot ($z = \lfloor \check{L}/2 \rfloor$). The out-of-plane spin components ($s_z \in [-1, 1]$) were systematically normalized to the strict interval $[0, 1]$ and mapped onto a Cartesian pixel grid, finalizing the input images $\mathbf{X}$ corresponding to the specific physical targets $\boldsymbol{\theta}$. The extended Hamiltonian parameter vector $\boldsymbol{\theta} \in \mathbb{R}^8$ estimated in this study is defined as: $\boldsymbol{\theta} = [T^{(0)}, \tilde{J}_2, \tilde{J}_3, \tilde{J}_4, \tilde{K}_{an1}, \tilde{K}_{anS}, \tilde{H}_{ex}, \tilde{K}_{DM}]$, where the first-shell exchange constant $\tilde{J}_1 = 1.0$ meV is fixed as the energy reference scale to break the scale symmetry of the inverse problem and ensure a well-posed regression target, as described in Section 3.1. Each of the eight free parameters was independently and uniformly sampled within the physically motivated bounds summarized in Table 1.

It is important to note that while all eight parameters are simultaneously estimated by the regression models, their effective identifiability from the s_z projection varies substantially due to the physical constraints of the observational modality. The higher-order exchange constants $\tilde{J}_3$ and $\tilde{J}_4$ produce energetic contributions that are strongly dominated by the first- and second-shell interactions $\tilde{J}_1$ and $\tilde{J}_2$, rendering their individual spatial signatures largely indistinguishable in the magnetization texture. Similarly, the magnetocrystalline anisotropy constants $\tilde{K}_{an1}$ and $\tilde{K}_{anS}$ and the external field $\tilde{H}_{ex}$ couple primarily to the in-plane spin components s_x and s_y, which are not observable in the s_z projection extracted from the central layer. These parameters were nevertheless included in the regression target $\boldsymbol{\theta}$ to provide a complete and physically honest characterization of the inverse problem, including its fundamental identifiability limits. Their inclusion does not degrade the predictive accuracy of the three most identifiable parameters ($T^{(0)}$, $\tilde{J}_2$, and $\tilde{K}_{DM}$), since the multi-output regression head learns to allocate gradient capacity independently to each output dimension. The resulting dataset is publicly available and can be accessed through the Kaggle repository[1]. It contains a total of 218,256 samples, where each input is a grayscale magnetic domain image of size 39×39×1 pixels, and each corresponding target is a vector of eight physical parameters $\boldsymbol{\theta} \in \mathbb{R}^8$ as defined above.

Table 1: Sampling ranges for the eight free Hamiltonian parameters constituting the regression target $\boldsymbol{\theta}$. The first-shell exchange $\tilde{J}_1 = 1.0$ meV is fixed as the energy reference. Parameters are sampled independently and uniformly within the specified bounds. Physical units: temperature in Kelvin [K], exchange and DMI constants in meV, anisotropy constants in meV/atom, external field in meV/atom.

Parameter	Physical role	Min	Max
$T^{(0)}$ [K]	Annealing temperature	0.2	20.0
$\tilde{J}_2$ [meV]	2nd-shell exchange	−0.66	0.66
$\tilde{J}_3$ [meV]	3rd-shell exchange	−0.29	0.29
$\tilde{J}_4$ [meV]	4th-shell exchange	−0.23	0.23
$\tilde{K}_{an1}$ [meV/atom]	Bulk anisotropy	0.0	0.2
$\tilde{K}_{anS}$ [meV/atom]	Surface anisotropy	0.0	0.2
$\tilde{H}_{ex}$ [meV/atom]	External Zeeman field	0.0	0.05
$\tilde{K}_{DM}$ [meV]	DMI strength	0.0	1.2

It is important to note that the DMI strength $\tilde{K}_{DM}$ is sampled exclusively over non-negative values ($\tilde{K}_{DM} \in [0.0, 1.2]$ meV). This restriction is a physically motivated convention arising from the symmetry properties of the antisymmetric exchange term in Equation ??. Under the transformation $\tilde{D} \to -\tilde{D}$, the equilibrium spin configuration undergoes a reversal of its in-plane chirality—the rotational sense of the chiral winding inverts—while the out-of-plane magnetization component s_z at every lattice site remains strictly invariant. Since the regression models operate exclusively on the two-dimensional s_z projection, configurations with $+\tilde{D}$ and $-\tilde{D}$ are exactly degenerate in the observation space, rendering the sign of the DMI fundamentally unidentifiable from s_z images alone. Including both signs simultaneously would introduce a strict two-fold degeneracy in the input–output mapping, collapsing the model's predictions toward $\hat{\tilde{D}} \approx 0$ and precluding meaningful regression. The chirality handedness is instead encoded through the fixed direction vector $\hat{\mathbf{d}}_{ij}$, consistent with standard conventions in atomistic modeling of interfacial DMI [53?]. Recovering the DMI sign would require access to chirality-sensitive observables, such as the full three-dimensional spin vector field or in-plane magnetization components, which constitutes a direction for future investigation (Section 6).

To ensure robust evaluation and prevent information leakage during the deep learning optimization phase, the generated dataset $\mathcal{D}$ is partitioned into training, validation, and testing subsets using a rigorous two-stage random split strategy. Initially, 15% of the total dataset is isolated as a hold-out test set to evaluate ultimate model generalization. Subsequently, the remaining 85% is further divided, allocating exactly 15% of the total initial data for validation. This procedure yields an effective split of 70% for training, 15% for validation, and 15% for testing. All partitions are executed utilizing a fixed random seed to guarantee strict reproducibility across all experimental runs. This strategy ensures that topological samples remain strictly independent across subsets while perfectly preserving the underlying thermodynamic distributions of the dataset.

Prior to network ingestion, both the input spatial magnetization images and the target Hamiltonian variables are preprocessed to ensure numerical stability and efficient gradient convergence. The physical target variables are scaled using Min-Max normalization, mapping each continuous parameter to the strict interval $[0, 1]$ based exclusively on the statistical distribution of the training set. This identical transformation is subsequently applied to the validation and test sets to maintain inferential consistency and directly support the bounded RAMs objective $O_p \in (0, 1]$. For the input images ($\mathbf{X}_n$), the single-channel s_z matrices are spatially resized to 224×224 pixels and duplicated across three channels (grayscale-to-RGB conversion). This dimensional expansion is critical to seamlessly leverage the hierarchical spatial filters learned from an ImageNet-based pretraining [64].

The configuration of the evaluated deep learning architectures (see Section 3.3) is summarized in Table 2. All four architectures share a unified preprocessing pipeline and an extended regression head designed to mitigate overfitting in the high-dimensional parameter space. Specifically, the global pooling output is passed through a Batch Normalization layer, followed by a Dropout layer with rate

[1] https://www.kaggle.com/datasets/carloscanamejoy/dataset-spines-complete

$p = 0.4$, a fully connected Dense layer of 256 units with ReLU activation and ℓ_2 weight regularization ($\lambda = 10^{-4}$), a second Batch Normalization layer, a second Dropout layer with rate $p = 0.3$, and a final linear Dense layer outputting the $\check{P} = 8$ target parameters. For the CNN-based architectures (ResNet50, DenseNet121, and Xception), the global aggregation is performed via Global Average Pooling over the spatial feature maps, whereas for the ViT-B/16 the token sequence output of the terminal encoder block is first collapsed via Global Average Pooling along the sequence dimension before entering the shared regularization head. This design choice reflects the distinct spatial inductive biases of convolutional and attention-based backbones while ensuring a strictly comparable regression capacity across all evaluated models.

The optimization of the parameter set $\hat{\Theta}$ was executed utilizing the Adam optimizer with a unified initial learning rate of $\eta = 10^{-4}$ for all architectures, including the ViT-B/16, reflecting the well-established sensitivity of both transformer-based and deep convolutional models to aggressive learning rate schedules when fine-tuning from ImageNet pretraining. All models were trained to minimize the MSE loss function, while the MAE was actively monitored as a secondary diagnostic. Training was conducted using a highly parallelized batch size of 512 samples over a maximum of 100 epochs. To actively prevent overfitting and manage the learning dynamics within the highly degenerate optimization landscape, two primary callbacks were implemented: a dynamic learning rate scheduler (ReduceLROnPlateau) that reduced the learning rate by a factor of 0.3 if the validation loss stagnated for 4 consecutive epochs, and an EarlyStopping protocol that halted training and restored the best network weights if no validation improvement was observed after 8 consecutive epochs.

The predictive performance on the unseen test set was quantitatively assessed using two complementary regression metrics. The coefficient of determination (R^2) quantifies the proportion of variance in each physical parameter successfully captured by the network:

$$R^2 = 1 - \frac{\sum_{n=1}^{N_{\text{test}}} \left(\theta_{n,p} - \hat{\theta}_{n,p}\right)^2}{\sum_{n=1}^{N_{\text{test}}} \left(\theta_{n,p} - \bar{\theta}_{p}\right)^2}, \tag{18}$$

where $\bar{\theta}_p$ is the mean of the observed ground truth target. Note that R^2 is statistically undefined when the target variable exhibits near-zero variance within a given evaluation subset; in these cases the metric is reported as N/A. The Mean Absolute Error (MAE) evaluates the absolute magnitude of the prediction error in the normalized feature space $[0, 1]$, providing a direct and interpretable measure of per-parameter accuracy:

$$\text{MAE} = \frac{1}{N_{\text{test}}} \sum_{n=1}^{N_{\text{test}}} \left|\theta_{n,p} - \hat{\theta}_{n,p}\right|. \tag{19}$$

Since all target variables are Min-Max normalized to $[0, 1]$ prior to training, the MAE is inherently dimensionless and directly comparable across all eight Hamiltonian parameters, irrespective of their original physical units or numerical ranges.

The RAMs were computed on the held-out test set using the formulations introduced in Section 3.4. For the Xception backbone, RAMs were extracted from eight layers selected to systematically cover the three functional stages of the architecture: two entry-flow layers capturing low-level spatial features at high resolution — block1_conv2 (112×112) and block3_sepconv2 (56×56) —; one transitional layer at block4_sepconv2 (28×28); five middle-flow layers operating at the bottleneck resolution of 14×14 — block6_sepconv2, block8_sepconv2, block10_sepconv2, and block12_sepconv2 —; and one exit-flow layer, block13_sepconv2 (14×14), immediately preceding the global pooling operation. This stratified selection was motivated by the physical hypothesis that early layers encode fine-grained domain periodicity and local spin-winding patterns at the scale of individual domain walls, while middle-flow layers progressively aggregate these local features into intermediate topological representations, and exit-flow layers encode high-level semantic information about the global magnetic phase. By distributing the selected layers across all three functional stages, the analysis directly tests whether the hierarchical decoupling predicted by the normalized RAM formulation of Equation 17 is empirically realized in the Xception backbone for Hamiltonian parameter estimation. For the ViT, a single attribution map per parameter was derived from the spatial token embeddings at the output of the terminal encoder block, reshaped into $\mathbf{A}_{\text{ViT}} \in \mathbb{R}^{14\times14\times768}$ following Equation 15. Since keras_hub.ViTBackbone does not expose internal token tensors as differentiable nodes within tf.GradientTape, the token sequence $\mathbf{Z}_s \in \mathbb{R}^{196\times768}$ was first extracted via a standard forward pass and then re-instantiated as a tf.Variable to enable gradient computation. The [CLS] token was treated as a fixed constant and the reconstructed sequence was passed through the GlobalAveragePooling1D and Dense regression head to compute O_p and its gradients with respect to $\mathbf{Z}_s$, following Equation 16.

The precision factor was fixed to $\gamma_p = 10$ uniformly for all $p \in \{1, \ldots, 8\}$. Given that all target variables are Min-Max normalized to $[0, 1]$, this value ensures a smooth and well-conditioned gradient signal across the full prediction error range: $O_p \geq e^{-10} \approx 4.5 \times 10^{-5}$ at maximum error and $O_p = 1$ at perfect prediction. Lower values of γ_p would yield spatially diffuse attributions insensitive to prediction quality, while higher values would cause gradient saturation for the less identifiable parameters where errors are systematically larger. The chosen value maintains sensitivity across all eight parameters, from the most identifiable ($T^{(0)}$, $\tilde{J}_2$, and $\tilde{K}_{DM}$) to the physically degenerate ones ($\tilde{K}_{an1}$, $\tilde{K}_{anS}$, $\tilde{H}_{ex}$).

To confine activations to the physical extent of the nanodot, a circular binary mask $\mathcal{M} \in \{0, 1\}^{224\times224}$ — defined as a disk of radius $r = 0.94 \times 112$ pixels centered at the

Table 2: Configuration of evaluated deep learning architectures for inverse Hamiltonian parameter estimation. All models share a standardized regularized regression head mapped to the $\check{P} = 8$-dimensional physical parameter space. BN: Batch Normalization; Drop(p): Dropout with rate p.

Model	Architecture Type	Pretraining	Global Aggregation	Output Head
ResNet50	Residual CNN	ImageNet	Global Avg. Pooling (spatial)	BN → Drop(0.4) → Dense(256, ReLU, ℓ_2) → BN → Drop(0.3) → Dense(8)
DenseNet121	Densely Connected CNN	ImageNet	Global Avg. Pooling (spatial)	BN → Drop(0.4) → Dense(256, ReLU, ℓ_2) → BN → Drop(0.3) → Dense(8)
Xception	Depthwise Separable CNN	ImageNet	Global Avg. Pooling (spatial)	BN → Drop(0.4) → Dense(256, ReLU, ℓ_2) → BN → Drop(0.3) → Dense(8)
ViT-B/16	Patch-based Transformer	ImageNet	Global Avg. Pooling (tokens)	BN → Drop(0.4) → Dense(256, ReLU, ℓ_2) → BN → Drop(0.3) → Dense(8)

image center — was applied to each RAM before normalization, setting all activations outside the disk to zero. A concentration score $\xi \in [0, 1]$, measuring the fraction of total activation energy within the disk, was used to select the most physically representative sample per magnetic phase: among the ten test images closest to the UMAP centroid of each phase cluster, the sample maximizing ξ averaged over all parameters and layers was chosen as the phase representative $\mathbf{X}_n^*$.

All experiments were conducted in a cloud-based environment using Google Colab. The computational setup was equipped with an NVIDIA H100 GPU with 97 GB of VRAM, supported by 210 GB of system RAM and 230 GB of available disk storage. The training pipeline was implemented in Python using TensorFlow, leveraging GPU acceleration for efficient model training and inference. The complete source code is publicly available at https://github.com/deathperminut/PaperInverseProblemEstimation, organized into three modules: /simulation containing the atomistic Monte Carlo Hamiltonian simulation scripts implemented in PyTorch with CUDA GPU parallelization; /regression containing the training, validation, and evaluation pipelines for all four deep learning architectures; and /rams containing the Regression Activation Maps computation and visualization code for both the Xception CNN and the Vision Transformer.

5. Results and Discussion

5.1. Thermodynamic Simulation of Magnetic Topologies

We present the main results of the proposed atomistic Hamiltonian simulation framework (see Section 3). Figure 7 delineates the empirical probability density functions—comprising histograms and kernel density estimates—for the extended parameter space θ, capturing the foundational statistical architecture of the simulated dataset. While the higher-order exchange constants ($\tilde{J}_2$, $\tilde{J}_3$, $\tilde{J}_4$) and the DMI ($\tilde{K}_{DM}$) exhibit tight, highly symmetric distributions, macroscopic scaling variables like the initial annealing temperature ($T^{(0)}$) display pronounced right-skewed behavior. This rigorous statistical characterization is crucial because these specific distributional shapes, bounded ranges, and sampling heterogeneities explicitly govern the topological diversity of the generated magnetic phase space. Consequently, this varying sampling density directly dictates the learning dynamics, optimization stability, and predictive limits of the downstream deep regression frameworks, particularly when navigating the highly degenerate regimes of the extended Heisenberg Hamiltonian.

To further validate the physical consistency of the generated dataset and provide an intuitive understanding of the underlying spin dynamics, Figure 8 illustrates the thermodynamic evolution of a representative magnetic nanodot during the Metropolis–Hastings Monte Carlo annealing process. This simulation is conducted under the fixed parameter configuration $\tilde{J}_1 = 1.0$ meV, $\tilde{J}_2 = \tilde{J}_3 = \tilde{J}_4 = 0.0$ meV, $\tilde{K}_{an1} = 0.10$ meV/atom, $\tilde{K}_{anS} = 0.00$ meV/atom, $\tilde{H}_{ex} = 0.00$ meV/atom, and $\tilde{K}_{DM} = 1.00$ meV, which constitutes a representative point within the full height-dimensional parameter space $\boldsymbol{\theta}$ defined in Table 1. The vanishing values of the higher-order exchange constants ($\tilde{J}_2$, $\tilde{J}_3$, $\tilde{J}_4$), surface anisotropy ($\tilde{K}_{anS}$), and external field ($\tilde{H}_{ex}$) isolate the dominant energetic competition between the first-shell symmetric exchange $\tilde{J}_1$ and the DMI $\tilde{K}_{DM}$, making this configuration particularly illustrative of the chiral ordering mechanism. The main panel tracks the total Hamiltonian energy $\mathcal{H}$ [meV] as a function of the exchange-normalized annealing temperature $T^{(0)}$ [K], while each inset displays the corresponding two-dimensional out-of-plane magnetization field s_z of the central layer at that specific thermal state. At elevated temperatures ($T^{(0)} \gtrsim 15$ K), the spin configurations exhibit a fully disordered, paramagnetic character, manifesting as spatially uncorrelated fluctuations of s_z with no preferred orientation. As the system is progressively cooled through the intermediate regime ($5\text{ K} \lesssim T^{(0)} \lesssim 15$ K), short-range correlated domains begin to nucleate, reflecting the growing dominance of the symmetric exchange interaction $\tilde{J}_1$ over thermal fluctuations. In this regime, the competition between $\tilde{J}_1$ and the $\tilde{K}_{DM}$ progressively induced non-collinear and chiral spin winding, visible as the emergence of alternating red-blue stripe-like patterns. Upon convergence to the near-ground state at $T^{(0)} = 0.1$ K, the system stabilizes into a well-defined labyrinthine domain configuration, characteristic of the helical phase governed by a strong DMI relative to the exchange energy. The monotonic decrease $\mathcal{H}$ throughout the annealing trajectory confirms the thermodynamic relaxation of the system toward a low-energy stable configuration, validating the physical correctness of the Monte Carlo pipeline and the quality of the generated dataset as input for the deep learning regression framework.

Figure 9 illustrates the converged ground-state spin configuration of a magnetic nanodot at $T^{(0)}_{final} = 0.1$ K,

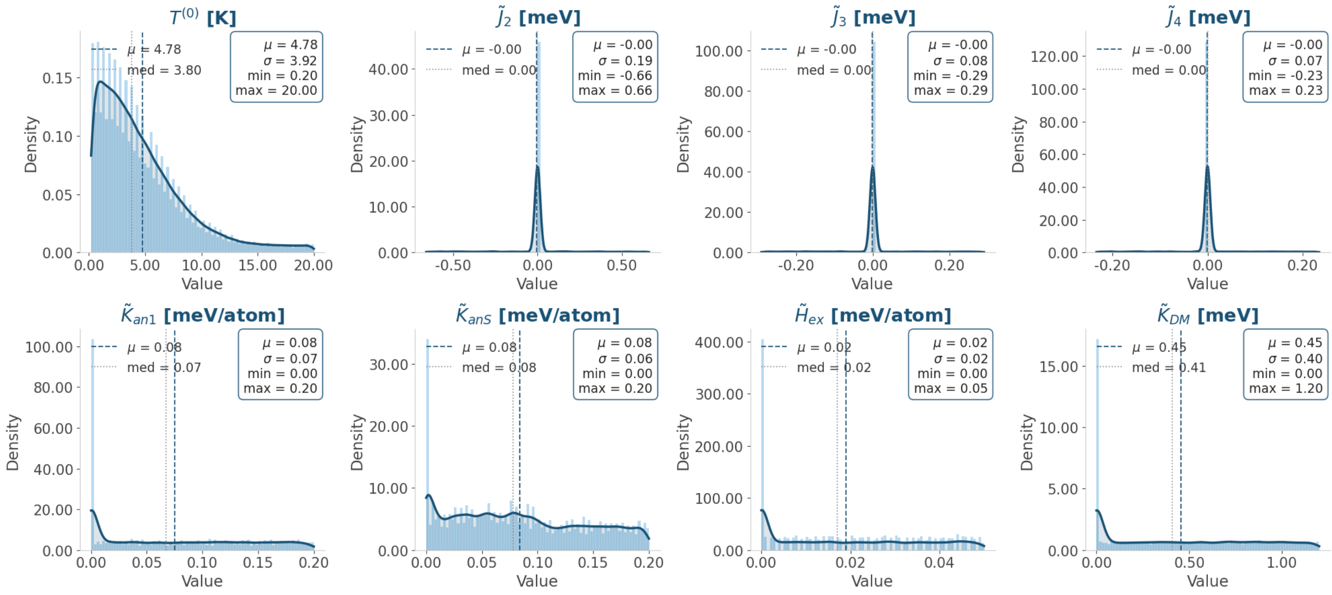

Figure 7: Empirical distributions of the thermodynamic and Hamiltonian parameters $\{T^{(0)}, \tilde{J}_2, \tilde{J}_3, \tilde{J}_4, \tilde{K}_{an1}, \tilde{K}_{anS}, \tilde{H}_{ex}, \tilde{K}_{DM}\}$ used to generate the dataset. Each subplot presents a histogram and KDE for a given variable, along with summary statistics including mean, standard deviation, and value range.

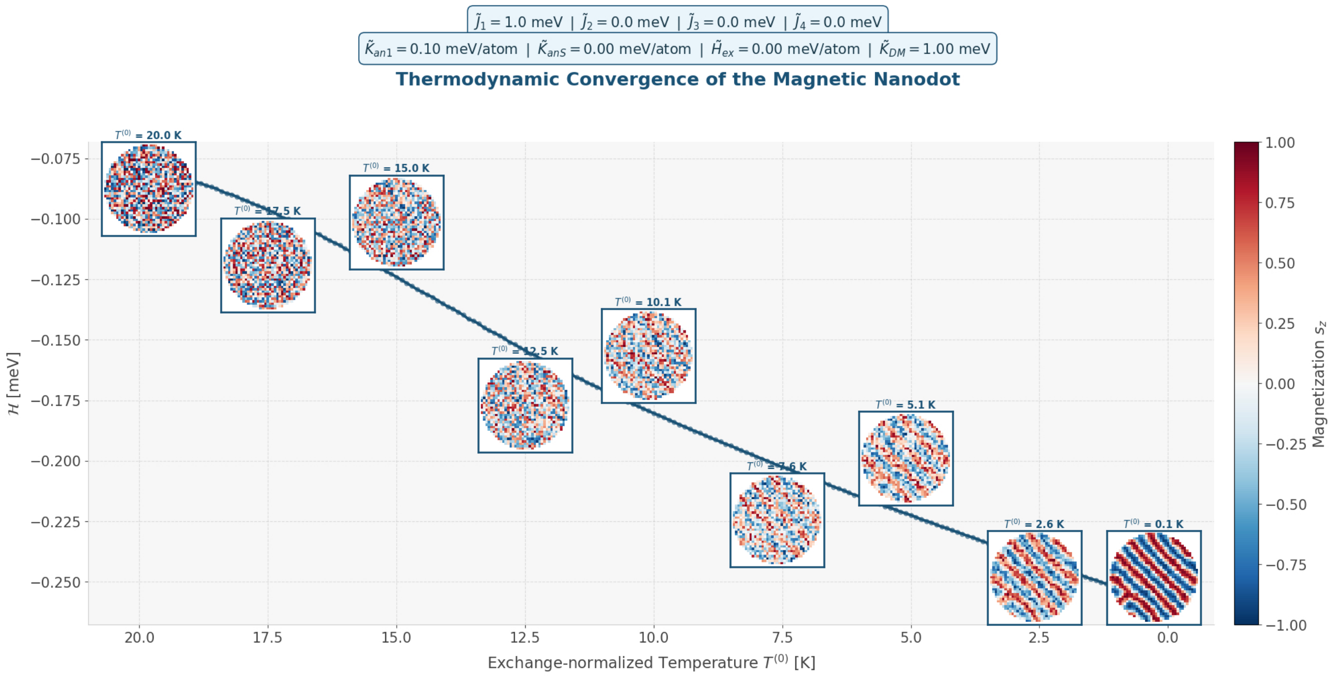

Figure 8: Thermodynamic convergence of a representative magnetic nanodot during simulated annealing. The main panel shows the total Hamiltonian energy $\mathcal{H}$ [meV] as a function of the exchange-normalized annealing temperature $T^{(0)}$ [K] under the fixed parameter configuration. Each inset depicts the two-dimensional out-of-plane magnetization field s_z of the central layer at the corresponding thermal state, illustrating the progressive transition from a fully disordered paramagnetic phase at high $T^{(0)}$ to a thermodynamically relaxed labyrinthine domain configuration at $T^{(0)} = 0.1$ K, driven by the competing energetic contributions of $\tilde{J}_1$ and $\tilde{K}_{DM}$.

simulated with specific fixed parameters ($\tilde{J}_1 = 1.0$ meV, $\tilde{K}_{DM} = 1.00$ meV, $\tilde{K}_{an1} = 0.10$ meV/atom, and all other parameters zeroed; see Table 1). Panel (a) maps the 2D out-of-plane magnetization (s_z) on the central layer—using dots for out-of-plane moments and arrows for transverse components—while Panel (b) displays the full 3D volumetric spin texture. The nanodot exhibits a labyrinthine domain pattern characterized by chiral walls. This morphology is driven directly by the energetic competition between the ferromagnetic symmetric exchange ($\tilde{J}_1$) and the DMI ($\tilde{K}_{DM}$). With a ratio of $\tilde{K}_{DM}/\tilde{J}_1 = 1.0$, the system stabilizes in the helical phase. By nullifying higher-order exchange, surface anisotropy, and external fields, this $\tilde{J}_1$–$\tilde{K}_{DM}$ competition is fully isolated, allowing the stripe-like domains to develop freely without preferential axial pinning.

Now, to fully characterize the structural diversity generated by the Monte Carlo annealing protocol across the entire dataset, Figure 10 presents two-dimensional phase diagrams of the magnetization image as a function of key Hamiltonian parameter pairs: $T^{(0)}$ versus $\tilde{J}_2$ and $T^{(0)}$ versus $\tilde{K}_{DM}$. These diagrams illustrate the macroscopic topological transitions driven by the underlying atomistic energetic competition. At elevated annealing temperatures, specifically above the effective Curie temperature T_C, thermal fluctuations overwhelm the magnetic interactions, collapsing all configurations into an indistinguishable, highly degenerate paramagnetic noise phase regardless of the specific exchange or DMI strengths. Conversely, in the low-temperature regime, the system stabilizes into ordered states, revealing distinct boundaries between chiral, ferromagnetic, and antiferromagnetic phases. Notably, the phase diagrams expose critical transition zones—such as the boundary between chiral and disordered states near $\tilde{K}_{DM} \approx 0$ meV, and the sign transition near $\tilde{J}_2 \approx 0$ meV—where disparate parameter combinations yield visually overlapping spatial textures. By comprehensively sampling this phase space, the thermodynamic simulation establishes a highly diverse and structurally ambiguous dataset (ill-conditioned mapping), setting a rigorously challenging foundation for the subsequent inverse deep learning parameter estimation.

5.2. Deep Learning Regression Results

The predictive performance of the studied architectures (DenseNet, ResNet, Xception, and ViT) was assessed on the held-out test set comprising 15% of the total dataset. Each model was tasked with simultaneously inferring the complete extended parameter vector $\boldsymbol{\theta}$ directly from the two-dimensional out-of-plane magnetization image $\mathbf{X}_n$, without any auxiliary physical information. As shown in Figures 11 and 12, no single architecture uniformly dominates both metrics: the ViT attains the highest R^2 score for seven of the eight Hamiltonian parameters ($T^{(0)}$, $\tilde{J}_3$, $\tilde{J}_4$, $\tilde{K}_{an1}$, $\tilde{K}_{anS}$, $\tilde{H}_{ex}$, and $\tilde{K}_{DM}$), while Xception achieves the highest R^2 for the second-shell exchange constant $\tilde{J}_2$. Notably, despite the ViT's superior variance-explained performance, Xception consistently produces the lowest Mean Absolute Error across most parameters, including $T^{(0)}$, $\tilde{J}_2$, $\tilde{J}_3$, $\tilde{J}_4$, $\tilde{H}_{ex}$, and $\tilde{K}_{DM}$, whereas the ViT only minimizes the MAE for the anisotropy constants $\tilde{K}_{an1}$ and $\tilde{K}_{anS}$. This complementary behavior between R^2 and MAE indicates that, while the ViT captures a larger fraction of the parameter variance and is therefore more sensitive to the long-range correlations governing global magnetic order Xception yields tighter absolute predictions in the dense ordered regions of the parameter space, where local spatial structure is the dominant predictive signal. The ViT's broad advantage in R^2 is physically interpretable: by processing the magnetization field $\mathbf{X}_n$ as a sequence of non-overlapping spatial patches and computing multi-head self-attention scores across all patch pairs, the ViT explicitly captures long-range correlations between distant spin textures, such as the global chirality imposed by $\tilde{K}_{DM}$ and the large-scale ordering driven by $\tilde{J}_1$, that are fundamentally inaccessible to purely local convolutional receptive fields. Xception's competitive behavior, in turn, stems from its depthwise separable convolution mechanism, which efficiently decouples the extraction of spatial spin-texture frequencies from the cross-channel correlations required to map those features onto the Hamiltonian parameter space, yielding the most accurate point predictions for the dominantly identifiable parameters.

Figures 11 and 12 reveal a pronounced and physically interpretable heterogeneity in predictive accuracy across the parameter space. The initial annealing temperature $T^{(0)}$, the second-shell exchange constant $\tilde{J}_2$, and the DMI strength $\tilde{K}_{DM}$ consistently yield the highest R^2 scores across all architectures, reflecting the robust and spatially distinctive topological imprints they leave on the observable magnetization texture: $T^{(0)}$ governs the global degree of spin order, $\tilde{J}_2$ controls the second-neighbor exchange pathway that modulates the domain-scale collinearity, and $\tilde{K}_{DM}$ directly dictates the characteristic wavelength and chirality of the domain patterns. However, a seemingly contradictory observation emerges when examining the MAE panels in Figure 12: despite their high R^2 scores, $T^{(0)}$, $\tilde{J}_2$, and $\tilde{K}_{DM}$ also exhibit comparatively larger absolute errors in their respective physical units — reaching up to approximately $1.0\,\mathrm{K}$ for $T^{(0)}$, $0.065\,\mathrm{meV}$ for $\tilde{J}_2$, and $0.05\,\mathrm{meV}$ for $\tilde{K}_{DM}$ in the worst-performing architecture. This apparent inconsistency is not a failure of the models but rather a direct consequence of the thermodynamic landscape of the dataset. As illustrated in the phase diagrams of Figure 10, the parameter space contains extensive high-temperature regions where thermal fluctuations above the effective Curie temperature T_C collapse the magnetization texture into an indistinguishable paramagnetic noise phase, regardless of the specific values of $\tilde{J}_2$ or $\tilde{K}_{DM}$. Within these thermally disordered regimes, visually identical configurations correspond to widely different parameter values, rendering the inverse mapping fundamentally ill-posed and inflating the absolute prediction error even when the global correlation structure — captured by R^2 — remains strong. The MAE values for $T^{(0)}$, $\tilde{J}_2$, and $\tilde{K}_{DM}$ therefore reflect the intrinsic degeneracy of the high-temperature paramagnetic phase rather than a systematic modeling deficiency: the same parameter that is most accurately recovered in the ordered low-

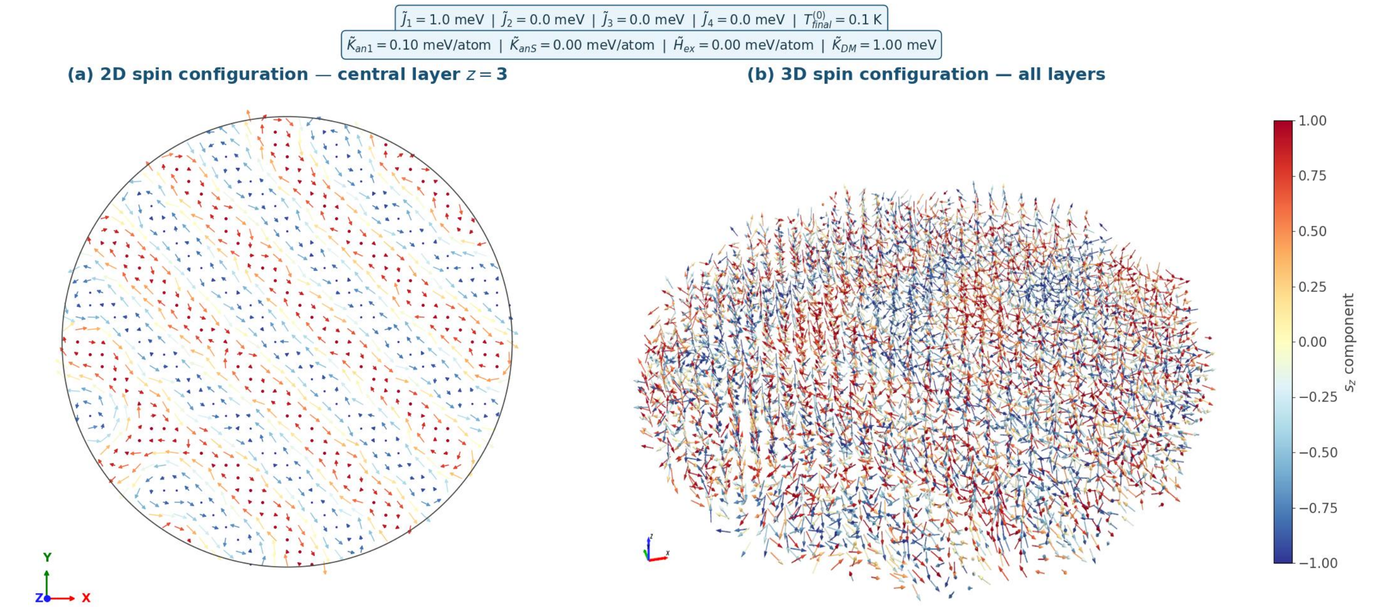

Figure 9: Converged ground-state spin configuration of a representative magnetic nanodot at $T^{(0)}_{final}$ = 0.1 K under a fixed parameter configuration. (a) Two-dimensional projection of the out-of-plane magnetization field s_z: colored dots indicate spins oriented predominantly out-of-plane, while colored arrows represent the in-plane orientation (s_x, s_y) of transverse moments. (b) Full three-dimensional spin configuration, revealing the volumetric coherence of the labyrinthine chiral domain structure stabilized by the competition between $\tilde{J}_1$ and $\tilde{K}_{DM}$. Color encodes the s_z component.

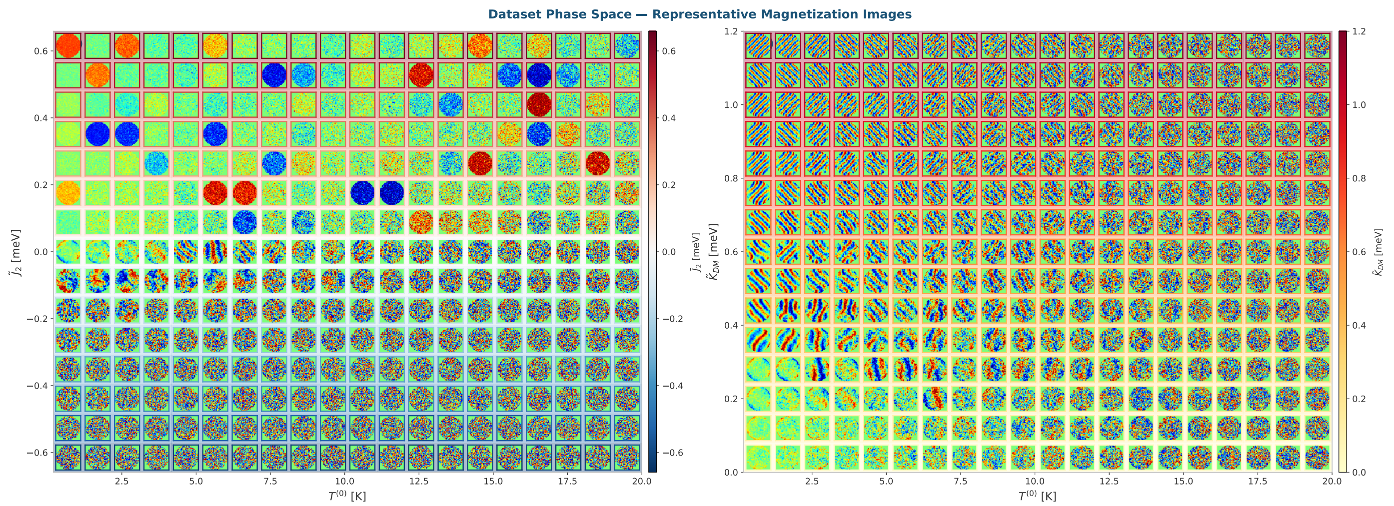

Figure 10: Two-dimensional phase diagrams of the magnetization texture $\mathbf{X}_n$ as a function of pairs of Hamiltonian parameters. (a) $T^{(0)}$ vs $\tilde{J}_2$. (b) $T^{(0)}$ vs $\tilde{K}_{DM}$. Each cell displays a representative magnetization image sampled from the dataset at the corresponding parameter combination; background color encodes the value of the vertical-axis parameter.

temperature regime (where the spin texture preserves a clear topological signature) is simultaneously the most degenerate in the disordered high-temperature regime (where the texture collapses into noise), producing non-negligible global MAE values driven primarily by the ill-conditioned subset of the dataset. A rigorous quantitative decomposition of this effect requires first separating the dataset into its constituent magnetic structural categories, as the global aggregation of all configurations inevitably conflates high-accuracy ordered-phase predictions with fundamentally ill-posed paramagnetic-regime estimates. This per-phase analysis is addressed further ahead, where the unsupervised structural clustering of the magnetization images allows us to isolate and quantify the specific contribution of each magnetic phase to the observed prediction error.

Conversely, the models exhibit two qualitatively distinct regimes of limited predictive performance for the remaining Hamiltonian parameters, which together delimit the fundamental identifiability boundaries of the inverse problem in the 2D observation space. The higher-order exchange constants $\tilde{J}_3$ and $\tilde{J}_4$ constitute the most severe case: their R^2 scores remain near zero across all evaluated architectures, indicating that the models are unable to recover meaningful information about these parameters beyond the dataset mean. Physically, the spatial signatures of these subordinate exchange constants are energetically dominated by the primary $\tilde{J}_1$ and $\tilde{J}_2$ interactions, progressively masking their observable effect on the magnetization texture and rendering them effectively unidentifiable from the s_z projection alone. Notably, however, their MAE values — on the order of 0.03–0.05 meV — remain numerically comparable to those of the better-identified parameters, which is a direct consequence of their narrow physical range in the dataset rather than genuine predictive skill: a model predicting the dataset mean for $\tilde{J}_3$ and $\tilde{J}_4$ will naturally incur a small absolute error simply because the dynamic range of these parameters is itself limited. The magnetocrystalline anisotropy constants $\tilde{K}_{an1}$ and $\tilde{K}_{anS}$ together with the external Zeeman field $\tilde{H}_{ex}$, in turn, occupy an intermediate identifiability regime. This partial predictability is physically consistent: these parameters couple primarily to the in-plane spin components s_x and s_y rather than to the observable out-of-plane component s_z, so the models are able to recover them only indirectly through the secondary topological imprints they induce on the chiral domain structure, such as the modulation of stripe periodicity or the preferential orientation of skyrmion cores. Furthermore, the visual manifestations of surface anisotropy $\tilde{K}_{anS}$ remain partially entangled with bulk anisotropy $\tilde{K}_{an1}$ and geometrically confined to the boundary layer of the nanodot, while the Zeeman contribution $\tilde{H}_{ex}$ — with MAE values below 0.009 meV/atom across all architectures — becomes increasingly indistinguishable from thermal disorder in the high-temperature regime. Together, these effects render the anisotropy and field parameters intrinsically partially observable and define the fundamental identifiability limits of the inverse problem within the two-dimensional s_z projection.

To gain deeper insight into the internal representations learned by the ViT during the inverse Hamiltonian estimation task, Figure 13 presents a Uniform Manifold Approximation and Projection (UMAP) of the model's latent space [65], where each point corresponds to a test sample colored by the value of each of the eight Hamiltonian parameters $\boldsymbol{\theta}$. The UMAP embedding was computed from the high-dimensional feature vectors extracted from the ViT's final encoder block prior to the regression head, providing a two-dimensional visualization of the geometric structure that the model has learned to impose on the magnetization image manifold. Rather than forming a single undifferentiated cloud, the latent space exhibits a rich topological structure composed of several geometrically distinct clusters and continuous gradients, reflecting the fact that the model has organized the magnetization textures according to their underlying physical regimes. Examining the coloring by $T^{(0)}$ reveals the clearest and most spatially coherent gradient across the embedding: low-temperature ordered configurations concentrate in well-separated compact regions, while high-temperature disordered states occupy a large, diffuse cluster in the upper portion of the latent space, consistent with the strong predictive performance of the model on $T^{(0)}$. The DMI strength $\tilde{K}_{DM}$ also produces a structured gradient in the latent space, with high-$\tilde{K}_{DM}$ helical configurations forming a distinct elongated region separated from the low-$\tilde{K}_{DM}$ disordered states, confirming that the ViT has successfully encoded the chiral winding imposed by the DMI as a geometrically meaningful direction in its latent representation. In contrast, the longer-range exchange constants $\tilde{J}_3$ and $\tilde{J}_4$ produce nearly uniform colorings across the entire embedding, indicating that the model's internal representations carry little discriminative information about these parameters a result that is fully consistent with their low R^2 values and their energetic subordination to the dominant first-shell exchange $\tilde{J}_1 = 1.0$ meV. Similarly, the anisotropy constants $\tilde{K}_{an1}$ and $\tilde{K}_{anS}$ and the external field $\tilde{H}_{ex}$ show spatially mixed and unstructured colorings, corroborating the partial observability limitations and the confounding effects identified in the regression analysis.

To further investigate the physical organization of the ViT's latent representations, a structured clustering analysis was conducted directly on the two-dimensional UMAP embedding of the test set feature vectors. Prior to visualization, the raw UMAP coordinates were normalized to the unit interval $[0, 1]$ along each axis using a min-max transformation. The spatial distribution of magnetization images across the normalized latent space, illustrated in Figure 14, reveals that the Vision Transformer has organized the magnetization textures into geometrically distinct and physically interpretable regions without any explicit supervision of magnetic phase labels. The lower-left region of the embedding is occupied by ordered, low-temperature spin configurations exhibiting well-defined helical stripe and labyrinthine domain patterns, whose spatial periodicity and chiral winding angle vary continuously along the diagonal axis of the manifold — a direct reflection of the systematic variation of $\tilde{K}_{DM}$ and $T^{(0)}$ within this phase. The upper central region concentrates thermally disordered configurations with increasing paramagnetic noise, consistent with samples at high $T^{(0)}$ where thermal fluctuations dominate over the competing energetic contributions of $\tilde{J}_1$ and $\tilde{K}_{DM}$. The isolated cluster visible in the upper-right corner of the embedding corresponds to a qualitatively distinct topological regime — ferromagnetic and skyrmion-like configurations with uniform or radially symmetric s_z distributions — whose geometric separation from the main manifold confirms that the Vision Transformer has successfully encoded these structures as a fundamentally different category of spin texture in its latent space.

To systematically identify and characterize the dominant magnetic phases encoded in the latent space, a

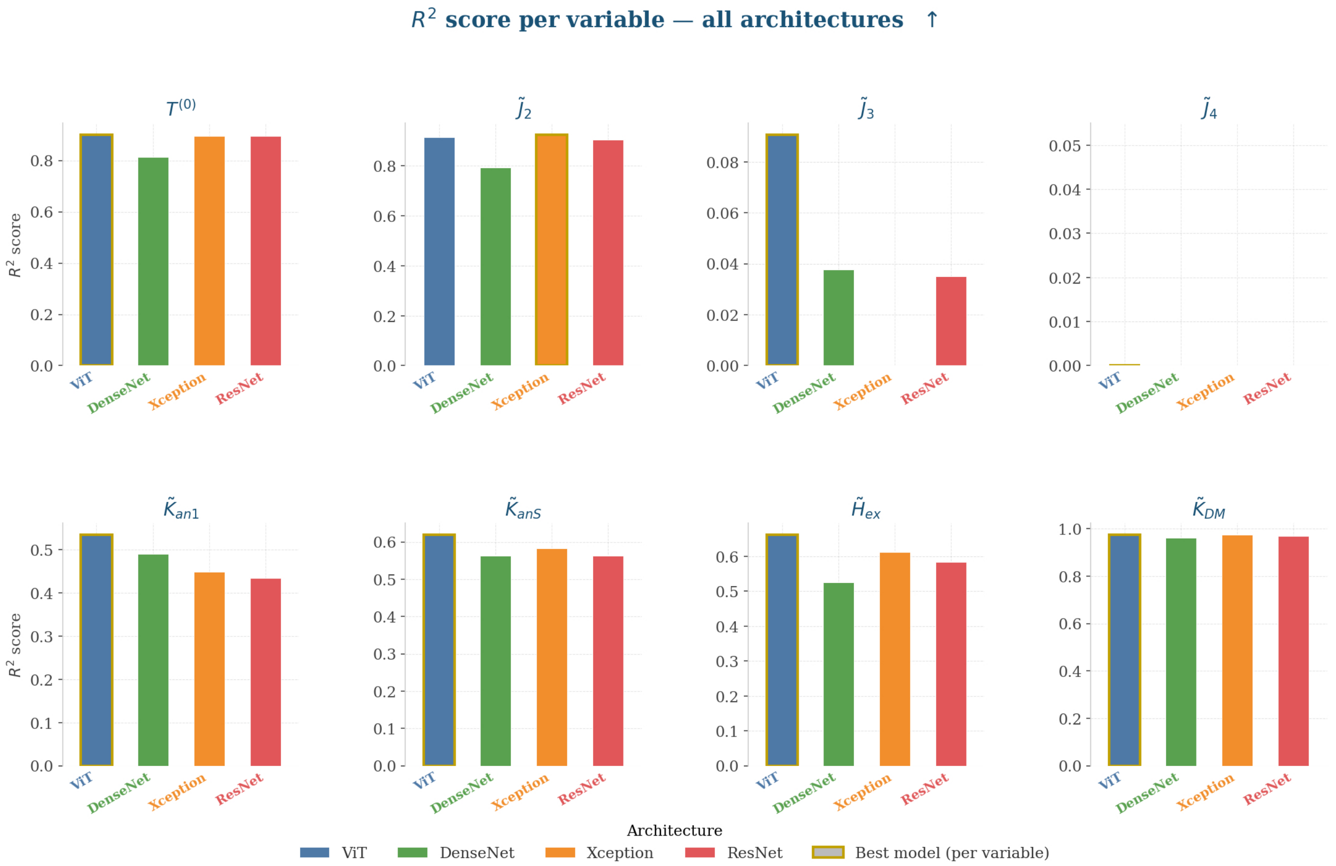

Figure 11: Per-variable R^2 score across all evaluated architectures for the eight Hamiltonian parameters $\boldsymbol{\theta}$, reported on the held-out test set. Each panel corresponds to one target parameter and shows one bar per architecture. Negative R^2 values, indicating predictions no better than the dataset mean, are clipped to zero for visualization clarity. Gold-bordered bars indicate the best-performing architecture for each parameter.

density-based clustering analysis was performed using the HDBSCAN algorithm on the UMAP embedding, followed by a KMeans subdivision of anomalously large clusters. Based on visual inspection of representative samples and the mean values of the Hamiltonian parameters $\boldsymbol{\theta}$ within each group, the resulting clusters were consolidated into four physically meaningful magnetic phase categories.

The ferromagnetic phase groups configurations characterized by near-uniform out-of-plane magnetization distributions, where the spins are predominantly aligned along a single direction with minimal spatial modulation, reflecting the dominance of ferromagnetic exchange interactions over chiral and thermal perturbations. The others category constitutes the most heterogeneous group in the dataset, gathering configurations that do not exhibit the defining spatial signatures of any single ordered magnetic phase. This category encompasses thermally disordered states in which thermal fluctuations have disrupted long-range spin order, transitional configurations located at the boundaries between competing phases, and topologically mixed textures whose spatial structure is ambiguous or intermediate between ordered regimes. The common characteristic of all configurations in this group is the absence of a distinctive and reproducible magnetization pattern, making them the most structurally ill-conditioned samples in the dataset. The labyrinthine & conical phase comprises configurations exhibiting well-defined, spatially extended stripe and labyrinthine domain patterns with clear periodicity and chiral winding, reflecting the stabilization of competing magnetic interactions into ordered mesoscopic textures at sufficiently low annealing temperatures. Finally, the helical phase contains the most spatially coherent and regularly ordered configurations in the dataset, displaying well-defined helical stripe patterns with consistent chiral winding direction and spatial periodicity across the entire nanodot, representing the most energetically ordered magnetic ground states accessible within the sampled parameter space.

Figure 15 presents five representative magnetization images per phase, selected as the samples closest to the centroid of each group in the UMAP embedding, along with the mean Hamiltonian parameters $\langle T^{(0)}\rangle$, $\langle \tilde{J}_2\rangle$, and $\langle \tilde{K}_{DM}\rangle$ characterizing each phase.

The per-phase regression performance of the Vision

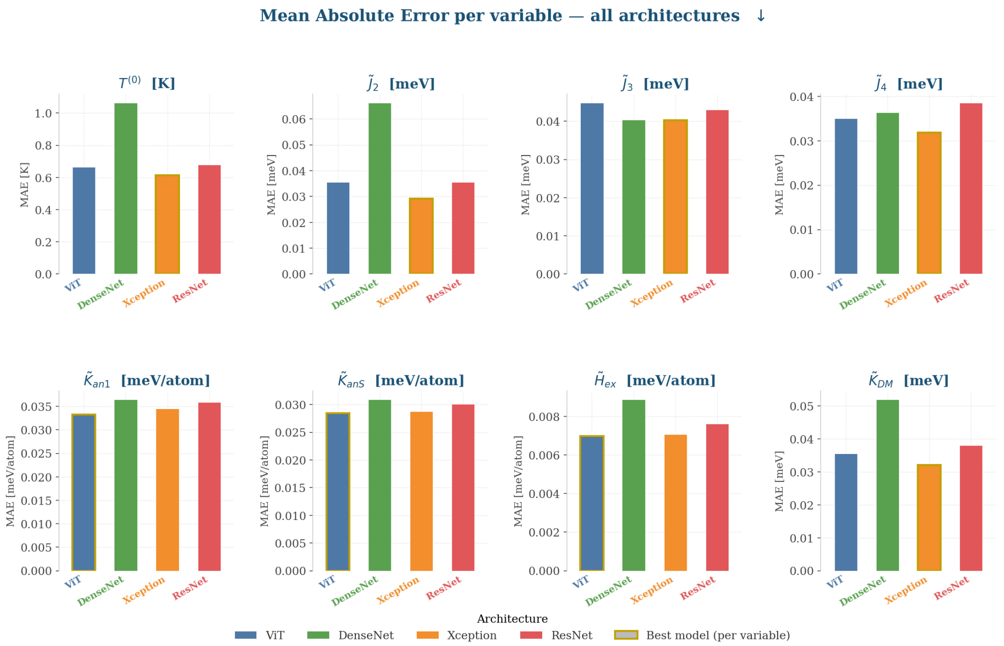

Figure 12: Per-variable Mean Absolute Error (MAE) across all evaluated architectures for the eight Hamiltonian parameters $\boldsymbol{\theta}$, reported on the held-out test set in the original physical units of each parameter (K for $T^{(0)}$, meV for exchange and DMI constants, and meV/atom for anisotropy and external field parameters). Each panel corresponds to one target parameter and shows one bar per architecture. The comparatively larger MAE observed for $T^{(0)}$, $\tilde{J}_2$, and $\tilde{K}_{DM}$ — reaching up to 1.0 K, 0.065 meV, and 0.05 meV respectively in the worst-performing architecture — is attributable to the thermally disordered paramagnetic configurations populating the high-temperature regime of the dataset (see Figure 10), where visually identical textures correspond to widely different parameter values, rendering the inverse mapping fundamentally ill-posed even when the global R^2 correlation remains strong. The comparatively small MAE of $\tilde{J}_3$ and $\tilde{J}_4$ despite their near-zero R^2 reflects the narrow dynamic range of these parameters in the dataset rather than genuine predictive skill. Gold-bordered bars indicate the best-performing architecture for each parameter.

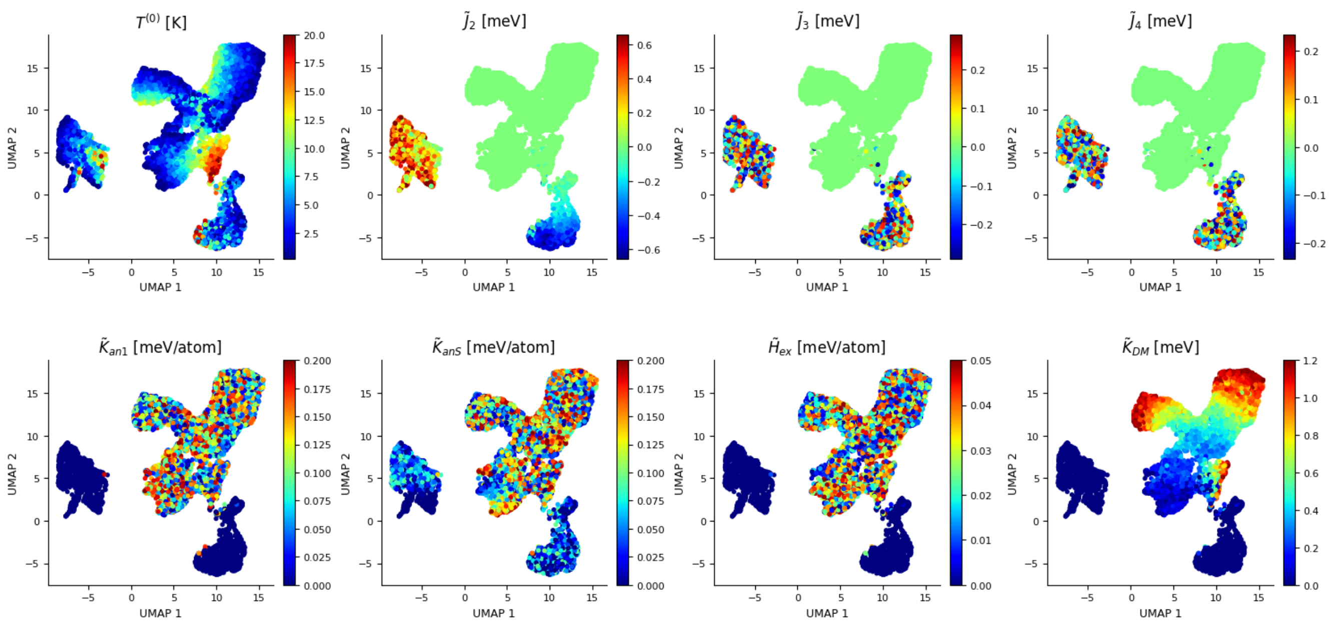

Figure 13: UMAP-based 2D projection of the ViT latent space on the held-out test set, colored by each of the eight Hamiltonian parameters $\boldsymbol{\theta} = [T^{(0)}, \tilde{J}_2, \tilde{J}_3, \tilde{J}_4, \tilde{K}_{an1}, \tilde{K}_{anS}, \tilde{H}_{ex}, \tilde{K}_{DM}]$. Each point represents a magnetization image $\mathbf{X}_n$ projected from the high-dimensional feature space of the final Transformer encoder block into two dimensions via UMAP.

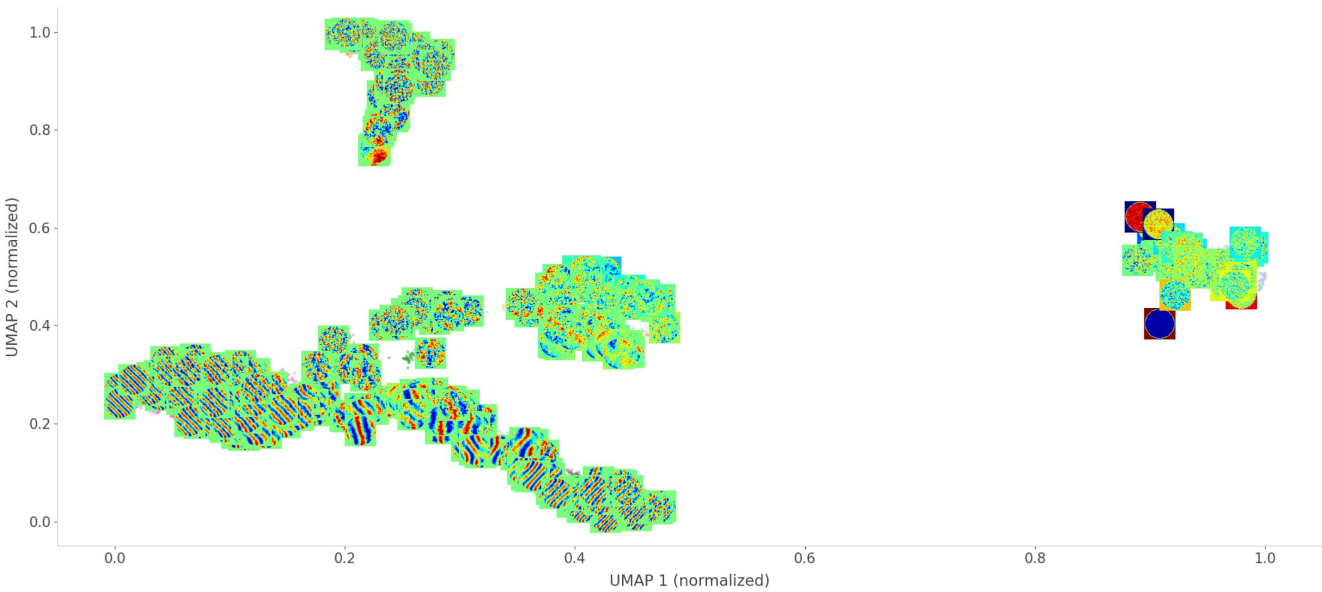

Figure 14: Spatial distribution of magnetization images $\mathbf{X}_n$ across the normalized two-dimensional UMAP projection of the ViT latent space (test set). Both axes are normalized to $[0, 1]$. Each thumbnail represents the out-of-plane magnetization field s_z of a randomly selected test sample placed at its corresponding normalized UMAP coordinate.

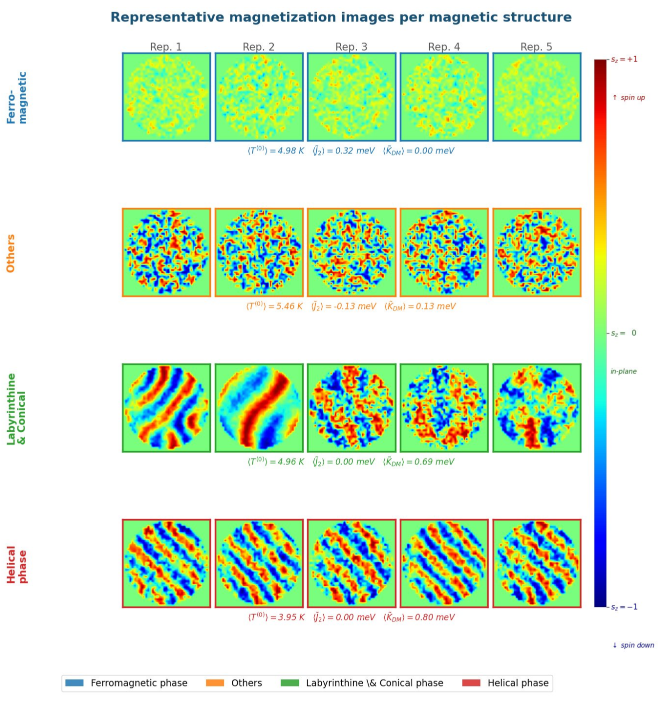

Figure 15: Representative magnetization images $\mathbf{X}_n$ for each of the four magnetic phase categories identified through clustering of the Vision Transformer latent space. Each row displays five samples selected as the nearest neighbors to the cluster centroid in the UMAP embedding. Mean Hamiltonian parameters $\langle T^{(0)} \rangle$, $\langle \tilde{J}_2 \rangle$, and $\langle \tilde{K}_{DM} \rangle$ are reported below each row. The Ferromagnetic phase displays near-uniform s_z distributions with minimal spatial modulation. The Others category gathers thermally disordered, transitional, and topologically mixed configurations that lack a defining spatial signature and represent the most structurally ambiguous samples in the dataset. The Labyrinthine & Conical phase exhibits spatially extended stripe and labyrinthine domain patterns with clear chiral winding. The Helical phase presents the most ordered and spatially coherent helical stripe configurations in the dataset, with consistent periodicity and chiral winding direction across the entire nanodot.

Transformer, evaluated on the 8,000 test samples used for the clustering analysis, is summarized in Figures 16 and 17. The results reveal a pronounced heterogeneity in predictive accuracy across magnetic phases that is physically consistent with the identifiability analysis presented earlier in this section. The labyrinthine & conical and helical phases achieve the highest R^2 values and the lowest MAE across all architectures, particularly for $T^{(0)}$ and $\tilde{K}_{DM}$, reflecting the fact that these ordered, low-temperature configurations exhibit the most discriminative and spatially coherent signatures in the s_z projection. Within these phases, the well-defined periodicity and chiral winding of the magnetization texture provide unambiguous observational constraints that allow the model to recover the underlying Hamiltonian parameters with high fidelity.

In stark contrast, the others category concentrates the largest absolute errors across virtually all Hamiltonian parameters, as evidenced by the dominant MAE values in Figure 17. This result directly confirms the physical interpretation advanced earlier: the thermally disordered, transitional, and topologically mixed configurations grouped in this category collapse into structurally indistinguishable magnetization textures despite corresponding to widely different parameter combinations, making the inverse mapping fundamentally ill-posed within this regime. The elevated MAE for $T^{(0)}$ in the others phase is particularly revealing, as it quantitatively isolates the contribution of the high-temperature paramagnetic regime to the global prediction error observed in the aggregated analysis. Similarly, the Ferromagnetic phase exhibits near-zero MAE for $\tilde{K}_{DM}$ and the exchange constants, consistent with the near-constant values of these parameters within this cluster and confirming that the apparent low error in these cases reflects degeneracy rather than genuine predictive capacity.

Within the ordered phases, the exchange constants $\tilde{J}_2$, $\tilde{J}_3$, and $\tilde{J}_4$ yield near-zero R^2 values for the Labyrinthine & Conical and Helical phases a direct consequence of the physical constraint that these textures are generated exclusively at $\tilde{J}_2 \approx 0$ meV, making the regression target degenerate and the inverse mapping statistically ill-posed for these parameters within these specific structural regimes.

5.3. RAMs-based Interpretability Results

To spatially trace the predictive behavior of the Xception architecture across the identified magnetic phase categories, RAMs were computed following the CNN-based formulation introduced in Section 3.4.1. For each of the four magnetic phases a single representative magnetization image $\mathbf{X}_n$ was selected as the sample closest to the centroid of the corresponding cluster group in the UMAP latent space, maximizing the concentration of gradient activations within the physical extent of the nanodot disk. RAMs were computed for the three most identifiable Hamiltonian parameters ($T^{(0)}$, $\tilde{J}_2$, and $\tilde{K}_{DM}$) across eight strategically selected convolutional layers of the Xception backbone, spanning the full depth of the network:

- block1_conv2: entry-level features at spatial resolution 112×112, capturing low-level pixel-scale magnetization gradients.
- block3_sepconv2 and block4_sepconv2: intermediate entry-flow layers at 56×56 and 28×28, encoding domain boundary frequencies and local spin-winding patterns.
- block6_sepconv2 through block12_sepconv2: middle-flow layers at 14×14, constituting the deepest repeated processing stage of Xception and encoding progressively more abstract topological representations of the magnetic texture.
- block13_sepconv2: exit-flow transition layer prior to global pooling, encoding high-level semantic representations of the global magnetic phase.

Selection of the precision factor γ.. The precision factor γ in the RAM objective $O_p = \exp\left(-\gamma\left(\theta_p - \hat{\theta}_p\right)^2\right)$ governs the selectivity of the gradient signal: small values render the objective nearly constant regardless of prediction quality, producing spatially uninformative maps, while excessively large values collapse the gradient to zero for parameters with non-negligible prediction error, yielding empty activation maps. To select γ in a principled and data-driven manner, a sensitivity analysis was conducted by evaluating the objective O_p at the empirical mean prediction error $\bar{\varepsilon}_p = \langle|\theta_p - \hat{\theta}_p|\rangle$ of each Hamiltonian parameter on the test set, for values spanning $\gamma \in \{0.1, 0.3, 0.5, 1, 2, 5, 10, 20, 50, 100\}$.

Figure 18 illustrates the resulting sensitivity curves. The left panel shows the shape of the Gaussian objective as a function of the prediction error ε for each candidate γ: values below $\gamma = 1$ produce flat curves that activate uniformly regardless of prediction quality, while values above $\gamma = 20$ generate overly steep curves that suppress the gradient for errors as small as $\varepsilon \approx 0.2$. The right panel evaluates O_p at the empirical mean error of each parameter: all variables maintain $O_p > 0.5$ for $\gamma \leq 10$, whereas for $\gamma \geq 20$ the parameters with larger prediction errors — specifically $T^{(0)}$ and $\tilde{K}_{DM}$, whose mean errors reach 0.17 and 0.24 respectively — begin to approach the gradient collapse threshold of $O_p < 0.05$. The value $\gamma = 10$ was therefore selected as the optimal operating point: it is the most selective value that simultaneously maintains a well-conditioned gradient signal for all eight Hamiltonian parameters, ensuring that the resulting RAMs are both spatially discriminative and physically interpretable across the full identifiability spectrum of the inverse problem.

To ensure that activation magnitudes remain physically interpretable and spatially confined to the nanodot geometry, each RAM was masked by a circular binary mask aligned with the disk boundary prior to normalization. The globally normalized RAMs $\tilde{\mathcal{R}}^{(l,p)}_{\mathrm{RAM}}$ were obtained following Equation 17, and the spatial patterns were additionally visualized under local per-cell normalization to re-

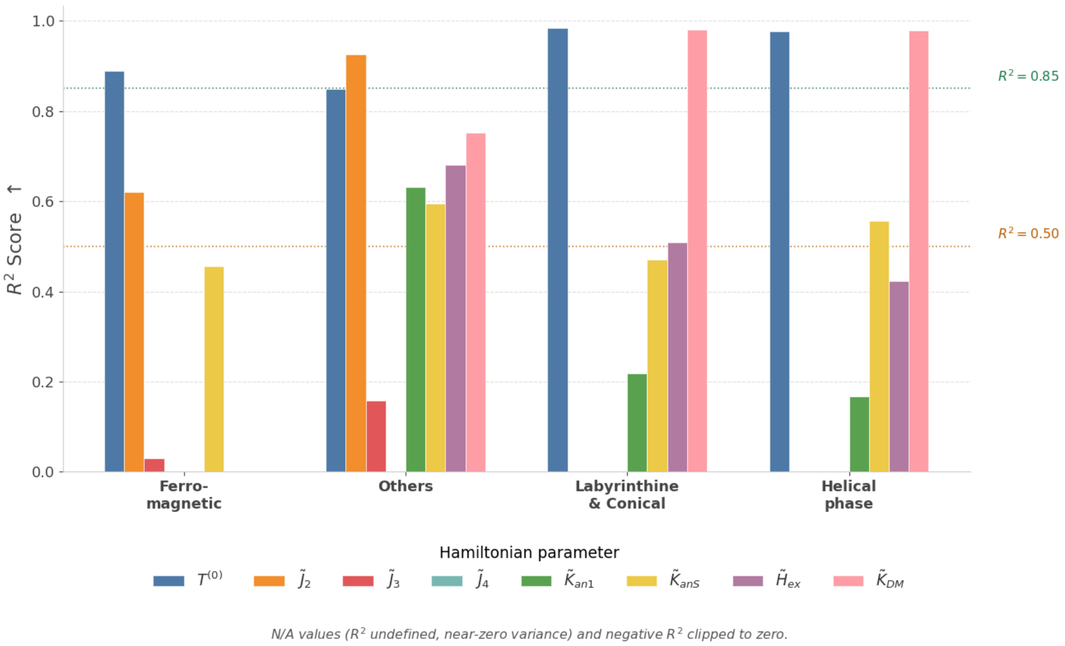

Figure 16: Per-phase R^2 score of the Vision Transformer across all eight Hamiltonian parameters $\boldsymbol{\theta}$ for the four identified magnetic phase categories (test set, $N = 8{,}000$). Each bar group corresponds to one magnetic phase and each bar to one target parameter. Dashed horizontal lines mark the identifiability thresholds at $R^2 = 0.85$ (high predictability) and $R^2 = 0.50$ (partial predictability). Negative R^2 values and near-zero variance cases (N/A) are clipped to zero. The Labyrinthine & Conical and Helical phases achieve the highest accuracy for $T^{(0)}$ and $\tilde{K}_{DM}$, while the Ferromagnetic phase exhibits the lowest identifiability due to the near-constant values of $\tilde{K}_{DM}$ and the exchange parameters within this topological regime.

veal the gradient attribution structure at each individual layer.

Figure 19 presents the resulting RAM overlays for the two most topologically structured phases — Helical and Labyrinthine & Conical — which were selected for visualization because their highly ordered, low-temperature spin textures produce the clearest and most physically interpretable gradient attribution patterns among the four computed phases, providing the strongest validation of the hierarchical decoupling hypothesis. The spatial attribution patterns reveal a physically consistent progression across network depth. In the early entry-flow layers (block1_conv2, block3_sepconv2), activations are spatially dense and highly structured, aligning with the fine-grained periodicity of the magnetization texture — particularly the stripe wavelength and domain wall boundaries that carry the primary signature of $\tilde{K}_{DM}$ and $T^{(0)}$. As depth increases through the middle-flow layers (block6 through block12), activations progressively consolidate into coarser, spatially localized hotspots that correspond to topologically significant regions of the nanodot — notably the chiral domain wall junctions and large-scale stripe boundaries characteristic of these two ordered phases. In the terminal exit-flow layer (block13_sepconv2), activations become highly concentrated in a small number of spatially compact regions, suggesting that the network's final semantic representations encode global topological stability rather than local texture periodicity. This hierarchical progression is consistent with the theoretical prediction of Section 3.4.3: early layers resolve the short-range energetic competition between $\tilde{J}_1$ and $\tilde{K}_{DM}$, while deeper layers integrate these local features into macro-scale thermodynamic representations governed by $T^{(0)}$ and the anisotropy constants.

To quantify the relative importance of each network layer and each Hamiltonian parameter in driving the gradient response, Figure 20 presents the mean maximum RAM activation per layer and variable, averaged over $N = 300$ randomly sampled test images per magnetic phase. To enable meaningful cross-layer and cross-parameter comparisons, the raw activation values were standardized via a global z-score transform across all layers and variables within each phase, and subsequently mapped through a sigmoid function $\sigma\left(\frac{x-\mu}{\sigma_x}\right) \in (0, 1)$, where μ and σ_x denote the global mean and standard deviation of the raw activations for each variable. This normalization preserves the relative ordering of activation magnitudes while compressing outliers and centering the scale around $\sigma = 0.5$, such that values above this threshold indicate above-average gradient response for that parameter at that layer depth.

This statistical analysis confirms and extends the qualitative observations from the spatial maps. Across all four phases, the sigmoid-normalized activation profile exhibits a non-monotonic behavior across depth, with no-

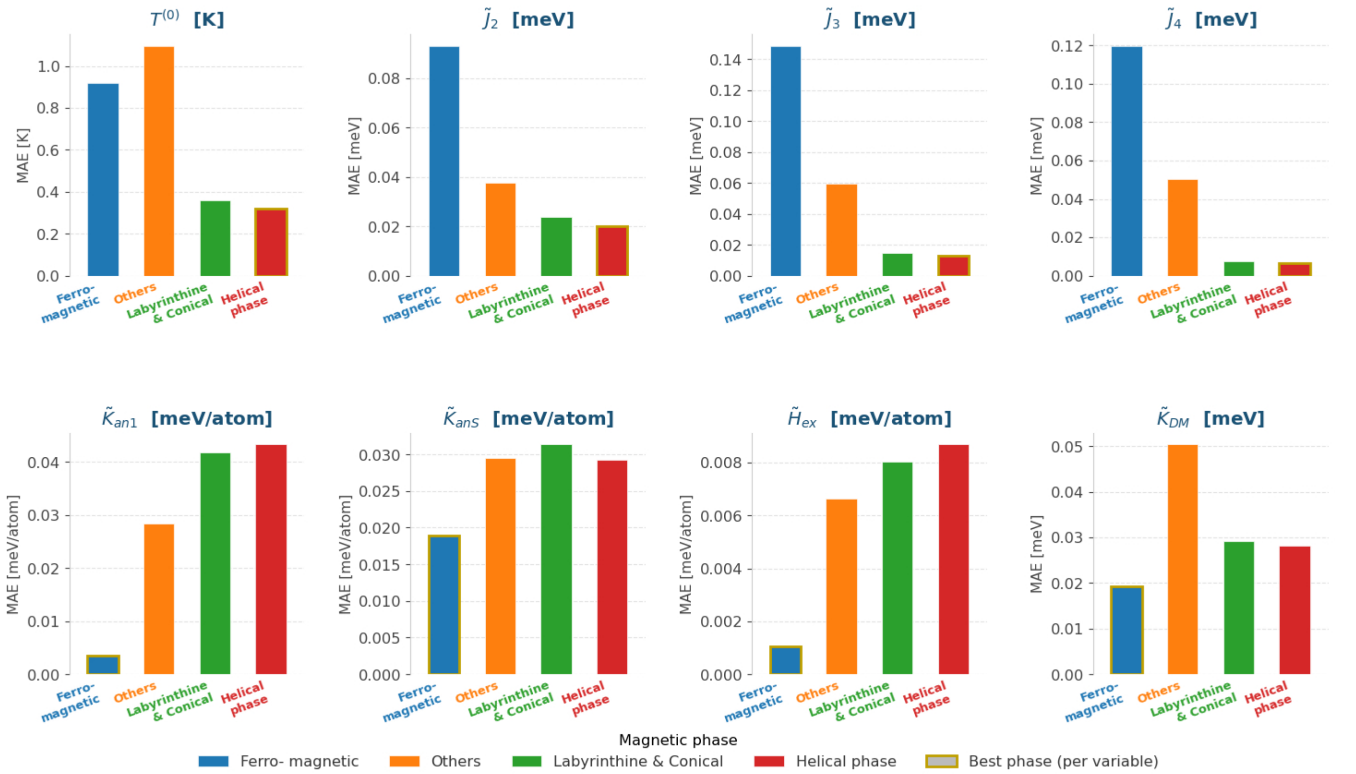

Figure 17: Per-phase Mean Absolute Error (MAE) of the Vision Transformer across all eight Hamiltonian parameters $\boldsymbol{\theta}$ for the four identified magnetic phase categories (test set, $N = 8{,}000$), reported in the original physical units of each parameter after inverse Min-Max scaling. Each panel corresponds to one target parameter and shows one bar per magnetic phase. The others category consistently concentrates the largest absolute errors across all parameters, confirming that the thermally disordered and transitional configurations within this group constitute the primary source of prediction error in the global regression analysis. Gold-bordered bars indicate the magnetic phase gur with the lowest MAE for each parameter.

table peaks at block8_sepconv2 and block13_sepconv2. The block8 peak is particularly pronounced for $\tilde{J}_3$ in the Others phase, suggesting that this middle-flow layer has learned to specifically encode the subtle energetic contributions of longer-range exchange interactions in the heterogeneous and topologically mixed configurations that characterize this category. The block13 peak, by contrast, is dominated by $\tilde{J}_2$ and $T^{(0)}$ across all phases, consistent with its role as a high-level semantic aggregator of the dominant physical parameters. The anisotropy constants $\tilde{K}_{an1}$ and $\tilde{K}_{anS}$ and the external field $\tilde{H}_{ex}$ systematically exhibit the lowest sigmoid activations across all layers and phases — consistently below $\sigma = 0.5$ — corroborating their limited identifiability established in Section 5.2 and confirming that the Xception backbone has not learned spatially distinctive features for these parameters, a result fully consistent with their physical coupling to the in-plane spin components s_x and s_y that are not observable in the s_z projection.

An analogous interpretability analysis was conducted for the Vision Transformer using the ViT-based RAM formulation introduced in Section 3.4.2. In contrast to Xception, which operates through a hierarchy of convolutional feature maps at varying spatial resolutions, the Vision Transformer processes the input magnetization image $\mathbf{X}_n$ as a sequence of $N_r = 196$ non-overlapping 16×16 patches, producing a pseudo-feature map $\mathbf{A}_{\mathrm{ViT}} \in \mathbb{R}^{14\times14\times768}$ upon token reshaping. Because the ViT architecture does not possess intermediate spatial hierarchies analogous to convolutional layers, the RAM analysis collapses to a single spatial attribution map per parameter, derived from the gradient of the error-decay objective O_p with respect to the spatial token embeddings $\mathbf{Z}_s$ at the output of the terminal encoder block.

Figure 21 presents the ViT RAM overlays for all four magnetic phases across the three most identifiable parameters. The spatial attribution patterns produced by the Vision Transformer are qualitatively distinct from those of Xception in two physically meaningful ways. First, the ViT RAMs exhibit smoother, patch-level spatial resolution consistent with the 14×14 token grid, capturing activation patterns at the scale of the magnetic domain wavelength rather than at the pixel level. This coarser but globally coherent attribution is a direct consequence of the self-attention mechanism, which explicitly correlates distant patches and therefore encodes long-range spin corre-

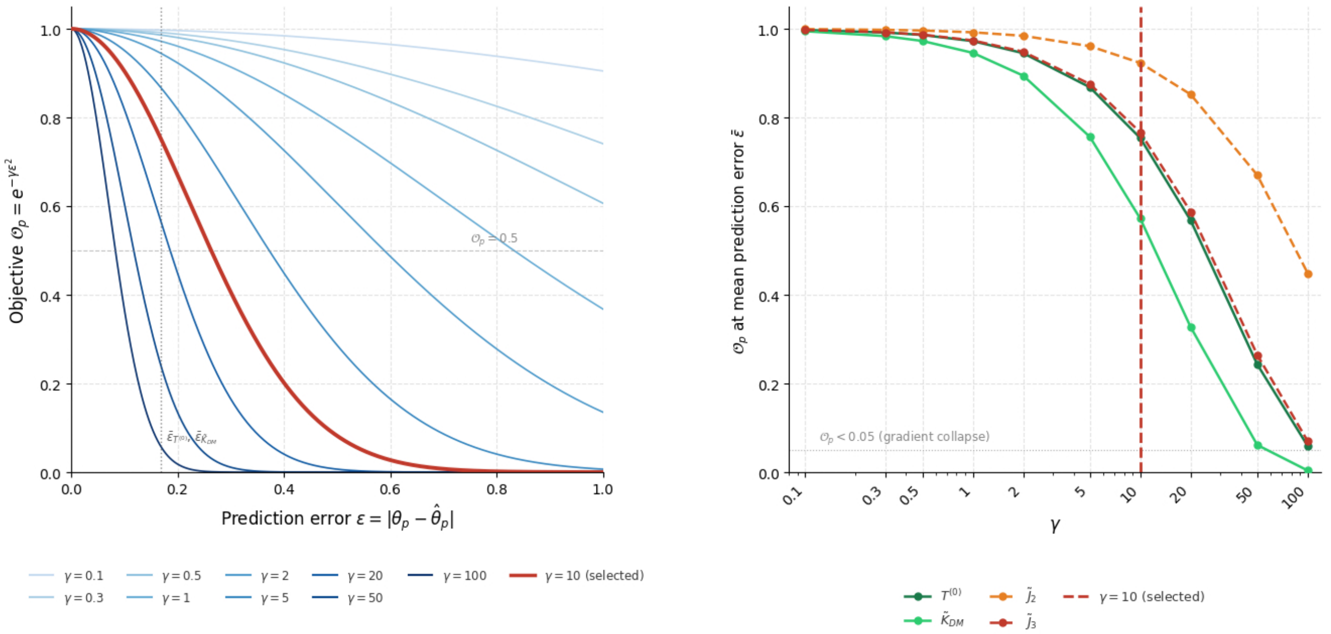

Figure 18: Sensitivity analysis of the RAM precision factor γ. Left: shape of the Gaussian objective $O_p = e^{-\gamma\varepsilon^2}$ as a function of the prediction error $\varepsilon = |\theta_p - \hat{\theta}_p|$ for $\gamma \in \{0.1, \dots, 100\}$; the selected value $\gamma = 10$ is highlighted in red. Vertical dotted lines indicate the empirical mean errors $\bar{\varepsilon}$ of $T^{(0)}$ and $\tilde{K}_{DM}$ on the test set. Right: O_p evaluated at $\bar{\varepsilon}_p$ for each Hamiltonian parameter as a function of γ. The horizontal dotted line marks the gradient collapse threshold $O_p = 0.05$. For $\gamma \leq 10$ all parameters maintain $O_p > 0.5$; for $\gamma \geq 20$ the parameters with larger prediction errors begin to approach the collapse threshold. The value $\gamma = 10$ is selected as the most selective operating point within the stable gradient regime.

lations that are physically relevant for parameters governing global magnetic order such as $T^{(0)}$ and $\tilde{K}_{DM}$. Second, the ViT RAMs display a stronger phase-specificity: for the Helical and Labyrinthine & Conical phases, activations concentrate along the chiral stripe boundaries and domain wall junctions — precisely the spatial features that encode the DMI-driven winding angle and the exchange-controlled domain spacing. Notably, the Others phase representative exhibits the highest annealing temperature in the dataset ($T^{(0)} = 17.00$ K), placing it well above the effective Curie temperature and producing a fully disordered paramagnetic texture; accordingly, the corresponding RAMs are spatially diffuse and structurally uninformative, reflecting the fundamental degeneracy of this regime. For the Ferromagnetic phase, the activations for $\tilde{K}_{DM}$ are notably sparse and peripheral, consistent with the near-zero DMI value ($\tilde{K}_{DM} = 0.00$ meV) of the representative sample and the model's inability to identify physically meaningful DMI features in the absence of chiral spin winding.

Figure 22 quantifies the mean maximum RAM activation per Hamiltonian parameter across all four magnetic phases for the Vision Transformer. Unlike the layer-resolved profile of Xception, the ViT activation spectrum reflects the integrated gradient response of the full token embedding space $\mathbf{Z}_s \in \mathbb{R}^{196\times768}$ for each parameter, with no layer hierarchy to resolve since the backbone produces a single sequence of spatial tokens. To enable meaningful cross-parameter and cross-phase comparisons, the raw activation values were standardized via a global z-score transform and subsequently mapped through a sigmoid function $\sigma\left(\frac{x-\mu}{\sigma_x}\right) \in (0, 1)$, where μ and σ_x denote the global mean and standard deviation of the raw activations for each variable across all phases. Values above $\sigma = 0.5$ indicate above-average gradient response for that parameter, providing a physically interpretable threshold for assessing relative parameter identifiability within the ViT token space.

The results reveal a physically consistent and phase-specific hierarchy of parameter identifiability. In the Ferromagnetic phase, $T^{(0)}$, $\tilde{J}_2$, and $\tilde{J}_3$ dominate the activation spectrum with sigmoid values well above $\sigma = 0.5$, reflecting the primacy of thermal and exchange-driven ordering as the primary discriminative features for this topologically simple configuration. The Others phase exhibits a more uniform and globally suppressed activation profile, consistent with the heterogeneous and thermally disordered nature of the configurations in this category, where no single parameter produces a spatially distinctive and consistent gradient signal across the ensemble of 300 samples. In the Labyrinthine & Conical phase, the activation spectrum shifts markedly toward the higher-order exchange constant $\tilde{J}_4$, the anisotropy parameters $\tilde{K}_{an1}$ and $\tilde{K}_{anS}$, the external field $\tilde{H}_{ex}$, and the DMI strength $\tilde{K}_{DM}$, all exhibiting sigmoid activations above $\sigma = 0.5$. This result suggests

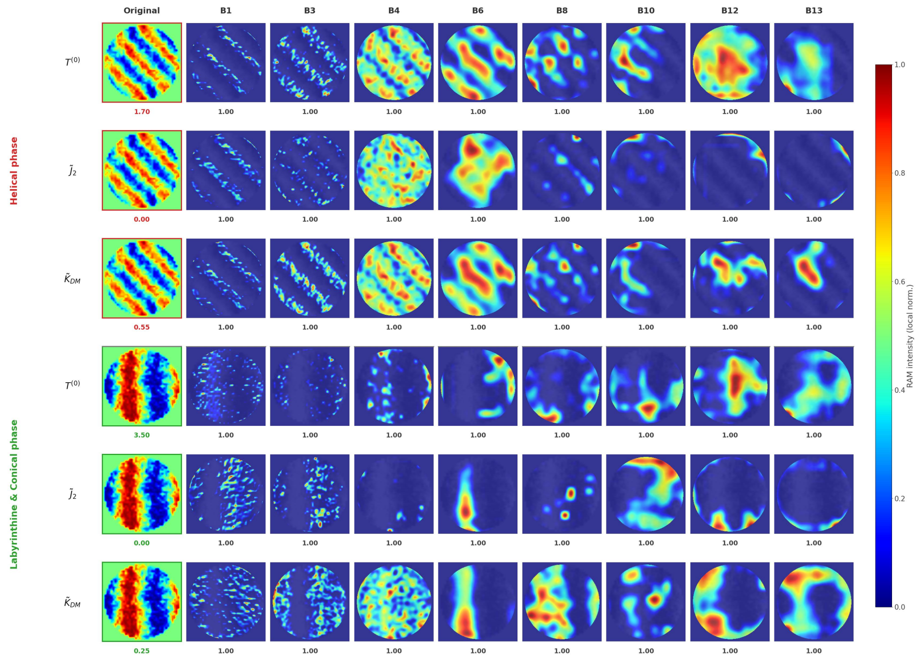

Figure 19: RAMs $\tilde{\mathcal{R}}^{(l,p)}_{\text{RAM}}$ for the Xception architecture, shown for the two most topologically structured magnetic phases — Helical and Labyrinthine & Conical — selected as the most representative cases from the full four-phase RAM computation. Three Hamiltonian parameters $\{T^{(0)}, \tilde{J}_2, \tilde{K}_{DM}\}$ are shown. Each panel displays the RAM overlay (jet colormap) on the original s_z magnetization image (grayscale), with local per-cell normalization to reveal spatial attribution patterns at each layer independently. Rows correspond to phase–parameter pairs; columns correspond to network layers from block1 (entry, 112×112) to block13 (exit, 14×14). The numerical score below each RAM cell reports the fraction of activation concentrated within the nanodot disk boundary, and the value below each original image reports the ground-truth parameter value for that representative sample. A physically consistent hierarchical decoupling is observed: early layers capture fine-grained domain periodicity while deeper layers consolidate activations into topologically significant spatial hotspots.

that the ViT has learned to resolve the energetic competition between chiral and anisotropic interactions that stabilizes the labyrinthine domain structure at low temperatures. Finally, in the Helical phase, $\tilde{K}_{anS}$, $\tilde{H}_{ex}$, and $\tilde{K}_{DM}$ clearly dominate the activation spectrum, consistent with the strong chiral winding and surface-symmetry-breaking effects that produce the spatially coherent helical stripe patterns characteristic of this phase. Across all phases, $\tilde{J}_3$ and $\tilde{J}_4$ maintain systematically low activations — with the exception of $\tilde{J}_4$ in the Labyrinthine & Conical phase — confirming that the ViT's global attention mechanism is generally unable to isolate spatially distinctive features for these energetically subordinate higher-order exchange constants. Taken together, the ViT RAM activation spectra provide a direct, architecture-agnostic validation of the identifiability hierarchy established through the regression analysis, demonstrating that the model's internal representations are physically grounded rather than statistically arbitrary.

Comparing the RAM-based interpretability results across Xception and the Vision Transformer reveals two complementary and physically consistent perspectives on how deep learning architectures resolve the inverse Hamiltonian estimation problem. Xception operates through a hierarchical spatial decomposition: early convolutional layers capture fine-grained domain periodicity and boundary chirality at high spatial resolution, while deeper middle-flow and exit-flow layers progressively consolidate these local features into coarser, topologically meaningful representations of the global magnetic phase. This hierarchical decoupling mirrors the multi-scale energetic competition of the extended Heisenberg Hamiltonian, where short-range exchange interactions ($\tilde{J}_1$, $\tilde{J}_2$) set the local domain scale while long-range DMI and thermodynamic effects ($\tilde{K}_{DM}$, $T^{(0)}$) govern global phase stability. The Vision Transformer, by contrast, resolves the same physical information through a single-scale, globally coherent patch-level attribution, leveraging multi-head self-attention to explicitly correlate distant spin textures across the nanodot. This architectural difference explains why the Vision Transformer achieves marginally superior overall regression performance — its global receptive field more efficiently captures the long-range topological coherence that distinguishes the identifiable parameters $T^{(0)}$ and $\tilde{K}_{DM}$ from the degenerate ones. In both architectures, the RAM analysis isolates the same set of physically meaningful spatial features as the primary drivers of accurate regression: chiral domain wall boundaries, stripe periodicity, and skyrmion-like core structures. Conversely, both architectures fail to produce spatially concentrated activations for $\tilde{K}_{an1}$, $\tilde{K}_{anS}$, and $\tilde{H}_{ex}$, confirming that the limited identifiability of these parameters is not a consequence of

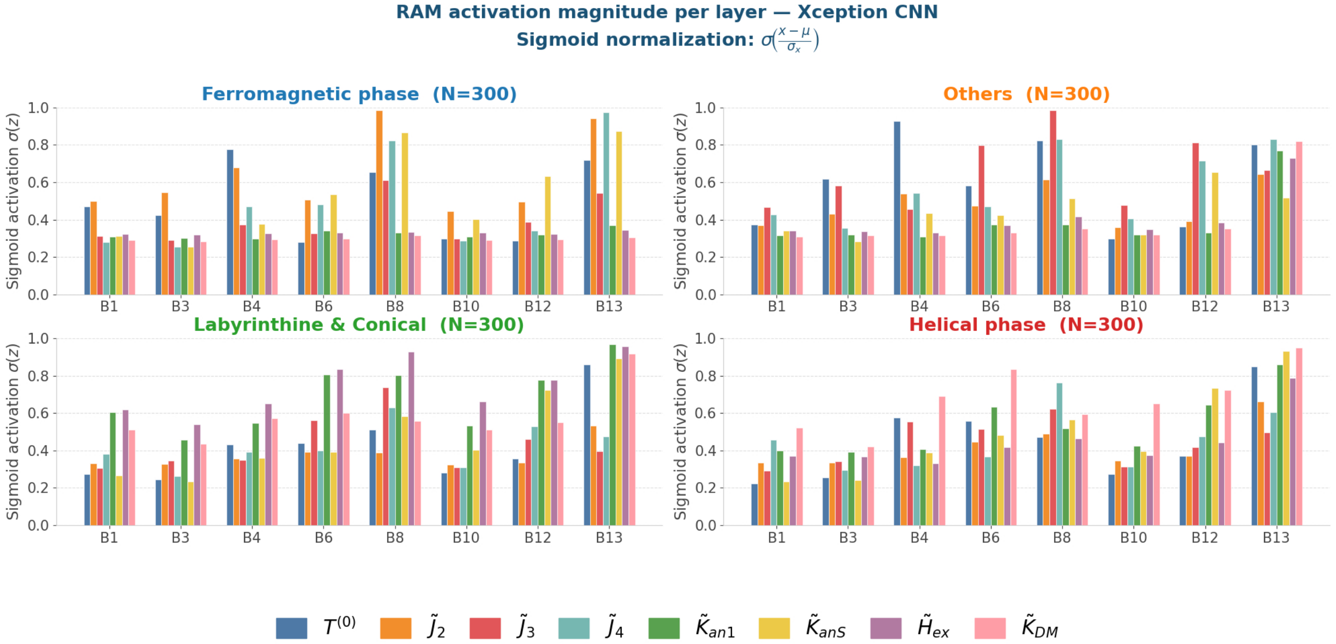

Figure 20: Mean maximum RAM activation per network layer and Hamiltonian parameter $\boldsymbol{\theta}$, averaged over $N = 300$ test samples per magnetic phase (Xception CNN). Activations are reported after sigmoid normalization $\sigma\left(\frac{x-\mu}{\sigma_x}\right)$ applied globally per variable, where values above $\sigma = 0.5$ indicate above-average gradient response. Each color corresponds to a distinct parameter following the nomenclature of Table 1. Activation is measured exclusively within the nanodot disk boundary (masked RAM). The non-monotonic depth profile with peaks at block8 and block13 reveals that the network resolves short-range exchange and DMI features in intermediate layers while aggregating global thermodynamic representations in the terminal exit-flow layer. Parameters with low identifiability ($\tilde{K}_{an1}$, $\tilde{K}_{anS}$, $\tilde{H}_{ex}$) consistently exhibit minimal sigmoid activation across all layers and phases, confirming the partial observability limitations identified in the regression analysis.

insufficient model capacity but of the fundamental physical constraints of the inverse problem — specifically, the partial observability of the anisotropy signal in the s_z projection and the confounding of the Zeeman field with thermal disorder in the two-dimensional magnetization image.

5.4. Comparative Analysis of Computational Inference Times

To comprehensively characterize the practical deployment viability of the proposed inverse Hamiltonian estimation framework, a systematic computational cost analysis was conducted across all four evaluated architectures, encompassing both the training phase and the inference regime. All experiments were executed on an NVIDIA H100 GPU (Google Colab Pro) with 97 GB of VRAM, using a unified training protocol of batch size $B = 512$ over a maximum of 100 epochs with 299 gradient steps per epoch, subject to early stopping with patience of 8 epochs on the validation loss.

Figure 23 presents the wall-clock training time per epoch and the per-step gradient update time for each architecture in the stable warm-start regime (epochs 2 onward, excluding the first epoch which incurs additional compilation and data-loading overhead). The results reveal a clear trade-off between model capacity and training efficiency. ResNet50 achieves the fastest training throughput at 65 s/epoch (217 ms/step), followed by DenseNet121 at 79 s/epoch (265 ms/step) and Xception at 103 s/epoch (343 ms/step). The Vision Transformer incurs the highest training cost at 167 s/epoch (558 ms/step) — a factor of 2.6× slower per epoch than ResNet50 and 1.6× slower than Xception.

This elevated training cost of the Vision Transformer is a direct consequence of its architectural complexity: with 85.8 million trainable parameters — 4.1× more than Xception (20.8M), 12.3× more than DenseNet121 (6.96M), and 16.7× more than ResNet50 (25.6M) — the self-attention mechanism requires substantially more floating-point operations per gradient step than the depthwise separable convolutions of the CNN baselines. Despite this overhead, the Vision Transformer's superior regression performance established in Section 5.2 demonstrates that this additional computational investment during training is justified by the quality of the learned Hamiltonian representations. The effective number of training epochs until early stopping convergence further modulates the total training wall-clock time: the Vision Transformer converged at epoch 68 while Xception converged at epoch 48, resulting in total training times of approximately 3.2 hours and 1.4 hours respectively.

Figure 24 presents the inference latency measurements for all four architectures across four batch sizes ($B \in$

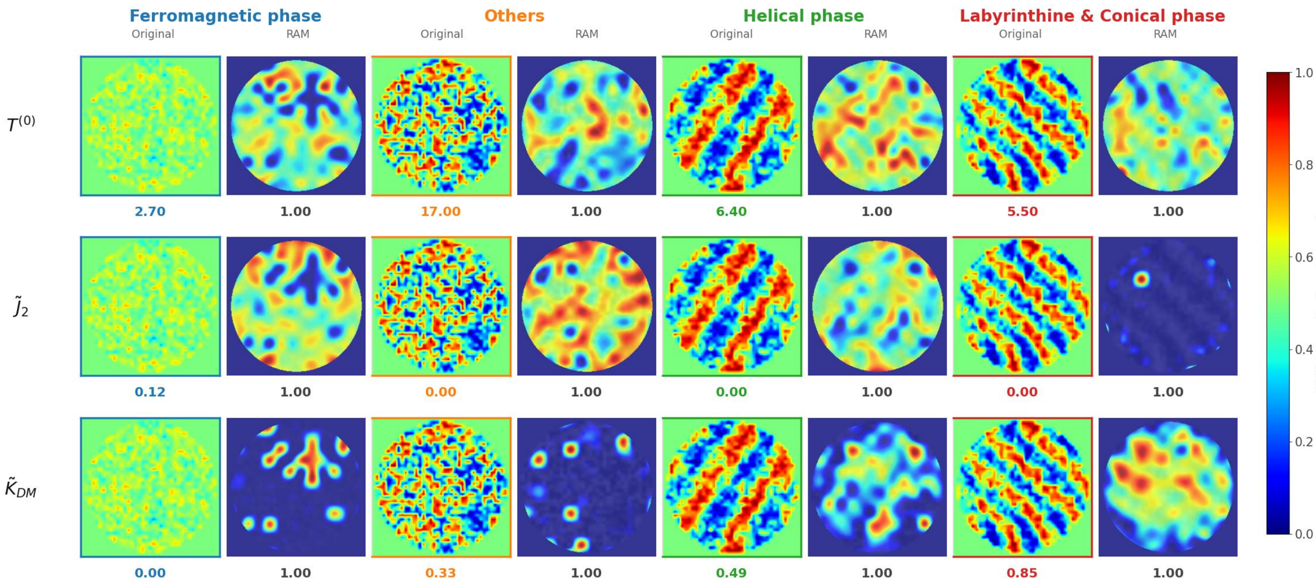

Figure 21: RAMs $\tilde{\mathcal{R}}^{(p)}_{\mathrm{RAM}}$ for the Vision Transformer across the four identified magnetic phases and three Hamiltonian parameters $\{T^{(0)}, \tilde{J}_2, \tilde{K}_{DM}\}$. Each panel pair displays the original s_z magnetization image (left) and the corresponding RAM overlay on grayscale (right), derived from gradients of the error-decay objective O_p with respect to the reshaped spatial token embeddings $\mathbf{A}_{\mathrm{ViT}} \in \mathbb{R}^{14\times14\times768}$. Score values report the fraction of activation within the nanodot boundary. The ground-truth parameter values reported below each original image confirm the physical regime of each representative: $\tilde{K}_{DM} = 0.00$ meV in the Ferromagnetic phase, $T^{(0)} = 17.00$ K in the Others phase reflecting a fully disordered paramagnetic configuration, and $\tilde{J}_2 = 0.00$ meV in both the Helical and Labyrinthine & Conical phases consistent with the energetic dominance of $\tilde{K}_{DM}$ in these chiral regimes. The ViT RAMs exhibit smooth, patch-scale spatial attribution with strong phase-specificity, concentrating on chiral domain walls and stripe boundaries in the ordered phases while producing spatially diffuse activations for the thermally degenerate Others phase.

$\{1, 8, 32, 128\}$), quantifying both the total batch processing time and the per-image throughput. Inference times are reported as mean $\pm$ standard deviation over $N = 10$ forward passes with 3 warmup passes excluded to eliminate GPU initialization effects.

The single-sample inference regime ($B = 1$) is the most relevant scenario for real-time physical characterization applications, where individual magnetization images are acquired sequentially and must be processed independently. In this regime, Xception achieves the lowest latency at 127.71 ± 1.46 ms/image, followed by the Vision Transformer at 188.89 ± 4.09 ms/image. DenseNet121 exhibits the highest single-image latency at 459.77 ± 7.22 ms/image — substantially slower than both architectures with significantly more parameters — a consequence of its densely connected skip connections that generate large intermediate feature tensors and cause memory bandwidth saturation on single-image workloads.

In the batch inference regime ($B = 128$), the per-image latency collapses dramatically across all architectures due to GPU parallelism. Xception again achieves the best throughput at 1.47 ± 0.007 ms/image, followed by the Vision Transformer at 1.92 ± 0.033 ms/image. This represents a 86.8× speedup for Xception and a 98.4× speedup for the Vision Transformer relative to their respective single-image latencies, demonstrating that both architectures are highly amenable to parallel batch processing in large-scale characterization pipelines. The throughput gap between Xception and the Vision Transformer narrows substantially at $B = 128$ (1.47 vs. 1.92 ms/image, a factor of only 1.31×), suggesting that the Vision Transformer's self-attention parallelism becomes increasingly competitive with convolutional processing at larger batch sizes.

Taken together, the computational analysis establishes a clear and physically motivated architecture selection criterion for the inverse Hamiltonian estimation task. For real-time single-image applications, Xception provides the optimal balance of inference speed (127.71 ms/image) and regression accuracy, ranking second overall in predictive performance. For high-throughput batch processing pipelines, the Vision Transformer is the preferred architecture: its superior predictive performance (Section 5.2) is achieved at a marginal throughput cost of 1.31× relative to Xception at $B = 128$, and its higher training cost is a one-time investment that need not be repeated at inference time. These results demonstrate that the proposed RAMs-based interpretability framework is computationally tractable for practical materials characterization scenarios, requiring sub-2 ms per image for both the best-performing architectures in the batch regime.

5.5. Limitations

Despite the promising results demonstrated across the regression, clustering, and interpretability analyses, the

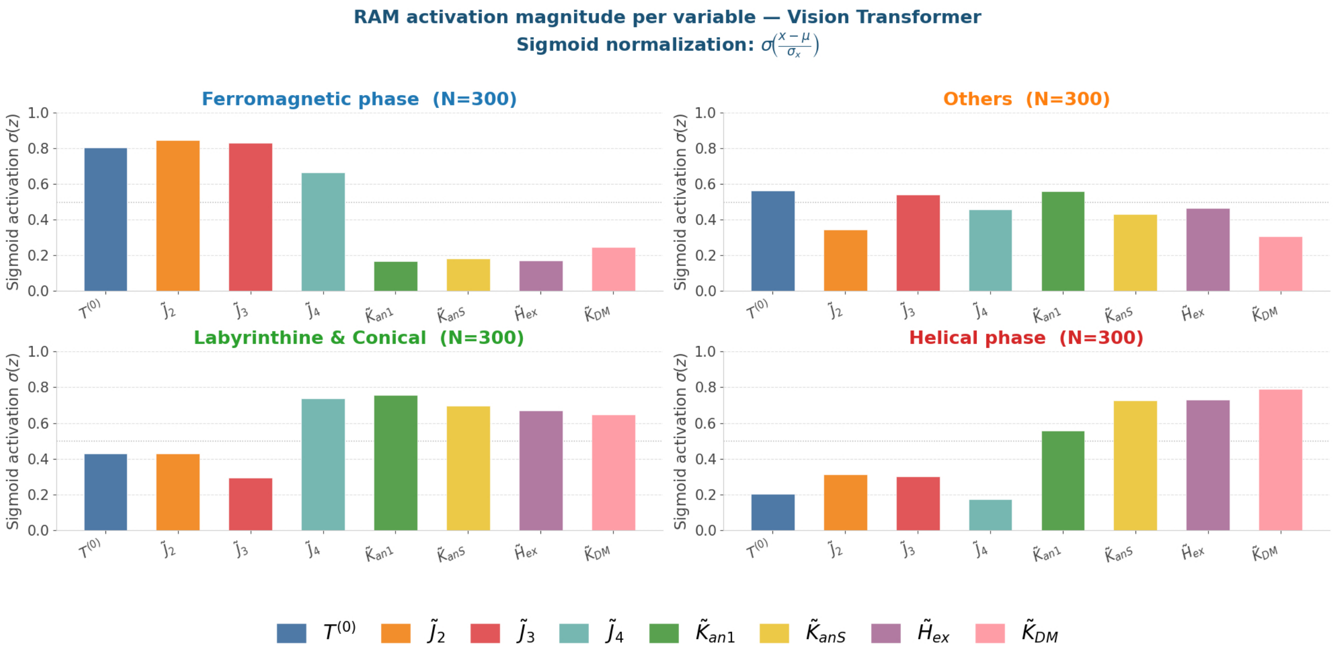

Figure 22: Mean maximum RAM activation per Hamiltonian parameter $\boldsymbol{\theta}$, averaged over $N = 300$ test samples per magnetic phase (Vision Transformer). Activations are reported after sigmoid normalization $\sigma\left(\frac{x-\mu}{\sigma_x}\right)$ applied globally per variable across all phases, where values above $\sigma = 0.5$ indicate above-average gradient response. Activation is measured within the nanodot disk boundary from gradients of O_p with respect to the reshaped spatial token embeddings $\mathbf{A}_{\text{ViT}} \in \mathbb{R}^{14\times14\times768}$ (Equation 15). Each bar color corresponds to a distinct parameter following the nomenclature of Table 1. The Ferromagnetic phase is dominated by $T^{(0)}$, $\tilde{J}_2$, and $\tilde{J}_3$; the Others phase shows a uniformly suppressed spectrum reflecting its thermally degenerate character; the Labyrinthine & Conical phase activates strongly on $\tilde{J}_4$, $\tilde{K}_{an1}$, $\tilde{K}_{anS}$, $\tilde{H}_{ex}$, and $\tilde{K}_{DM}$; and the Helical phase is governed by $\tilde{K}_{anS}$, $\tilde{H}_{ex}$, and $\tilde{K}_{DM}$, corroborating the identifiability analysis of Section 5.2.

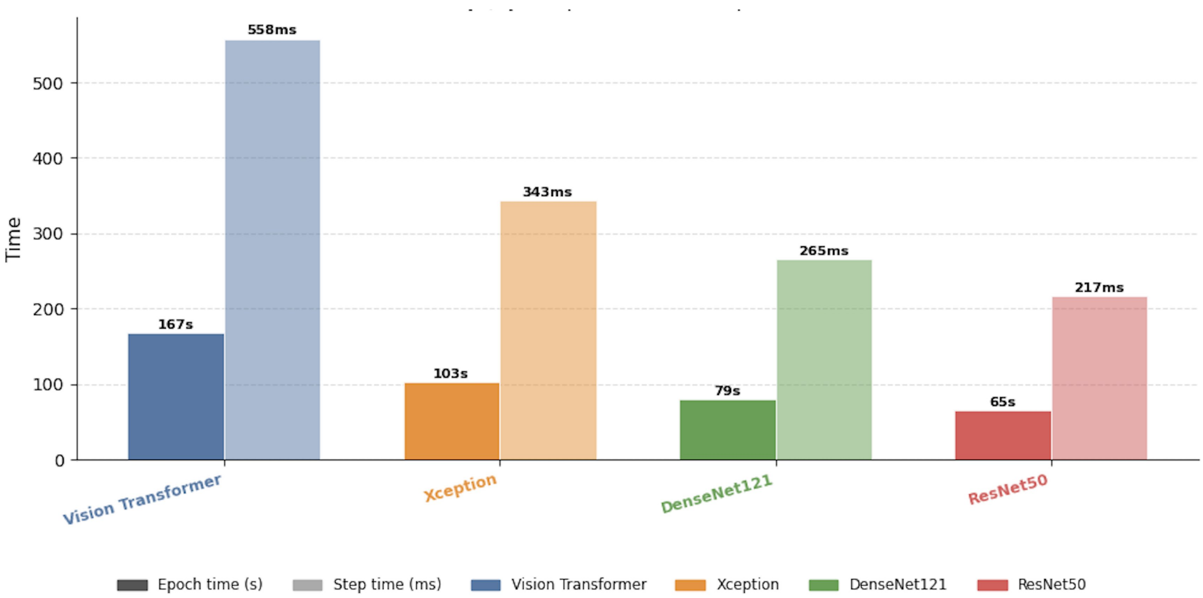

Figure 23: Training computational cost comparison across the four regression architectures on an NVIDIA H100 GPU (Google Colab Pro). (a) Wall-clock time per epoch in the stable warm-start regime. (b) Per-step gradient update time (299 steps/epoch, batch size $B = 512$). The Vision Transformer incurs the highest training cost (167 s/epoch, 558 ms/step) due to its 85.8M-parameter self-attention architecture, while ResNet50 achieves the fastest throughput (65 s/epoch, 217 ms/step). The training cost hierarchy directly reflects the architectural complexity of each model.

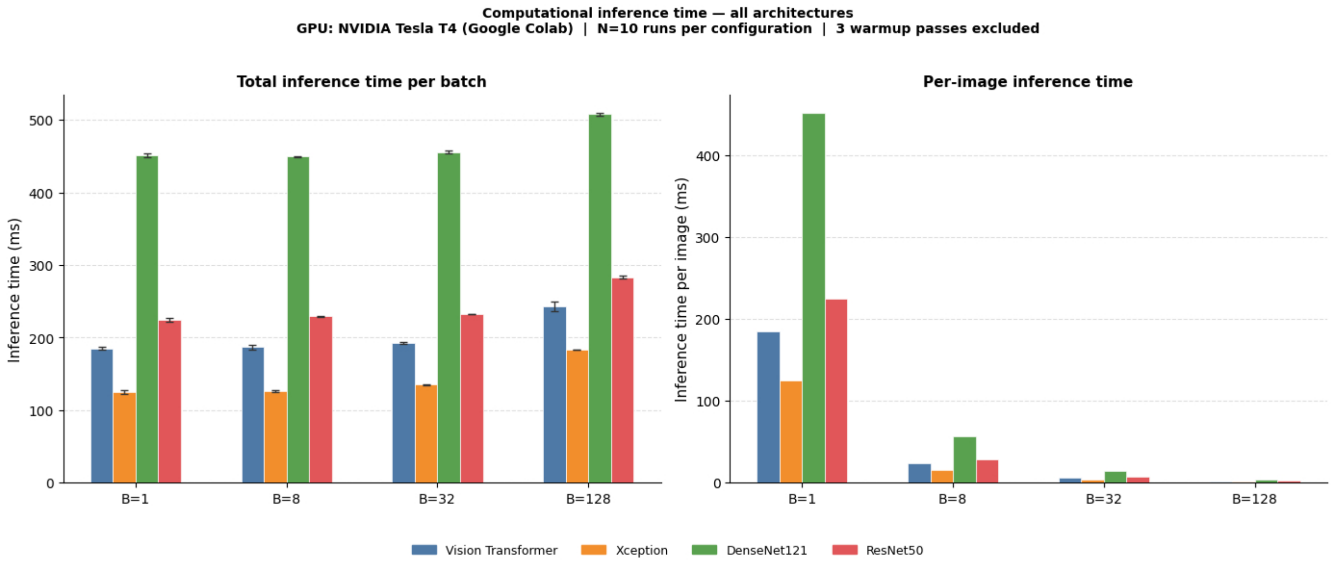

Figure 24: Inference latency comparison across the four regression architectures on an NVIDIA H100 GPU (Google Colab Pro) for batch sizes $B \in \{1, 8, 32, 128\}$. Each bar reports the mean ± standard deviation over $N = 10$ forward passes (3 warmup passes excluded). (a) Total batch processing latency in milliseconds. (b) Per-image latency in ms/image, illustrating the amortization of GPU overhead across parallel samples. Xception achieves the lowest per-image latency in both the single-sample (127.71 ms) and batch (1.47 ms) regimes. The Vision Transformer exhibits competitive batch throughput at $B = 128$ (1.92 ms/image), with a factor of only 1.31× overhead relative to Xception, despite its 4.1× larger parameter count.

proposed framework presents several inherent limitations that must be explicitly acknowledged to contextualize its applicability and guide future research directions.

The most fundamental limitation of the framework is the intrinsic partial observability of the extended Heisenberg Hamiltonian from two-dimensional magnetization images. As established in Section 5.2 and confirmed by the RAM-based interpretability analysis of Section 5.3, the projection of the three-dimensional spin configuration onto the out-of-plane component s_z of the central layer of the nanodot constitutes a dimensionality reduction that irreversibly discards physical information. The magnetocrystalline anisotropy constants $\tilde{K}_{an1}$ and $\tilde{K}_{anS}$, the external Zeeman field $\tilde{H}_{ex}$, and the higher-order exchange constants $\tilde{J}_3$ and $\tilde{J}_4$ produce spatially degenerate magnetization signatures in the s_z projection that cannot be fully disentangled by any purely image-based regression model, regardless of its capacity or architecture. Two distinct observational constraints contribute to this limitation. First, the anisotropy and Zeeman parameters couple primarily to the in-plane spin components s_x and s_y rather than to s_z, so they are recoverable only indirectly through the secondary topological imprints they induce on the chiral domain structure. Second, the input images are extracted exclusively from the central discrete layer ($z = \lfloor \check{L}/2 \rfloor$) of the three-dimensional nanodot lattice, discarding the depth-resolved spin structure that encodes the surface-to-bulk anisotropy ratio $\tilde{K}_{anS}/\tilde{K}_{an1}$ and the conical tilt angle of helical spin textures. Recovering these parameters with high fidelity would therefore require either access to the full three-dimensional spin vector field $\mathbf{S}_i = (s_x, s_y, s_z)$ at each lattice site, or volumetric magnetization inputs obtained through multi-layer or tomographic magnetic imaging techniques. This is not a limitation of the deep learning models themselves but of the observational modality assumed by the framework, which mimics the two-dimensional projection inherent to most experimental magnetic imaging techniques such as magnetic force microscopy or Lorentz transmission electron microscopy.

The dataset utilized in this study consists entirely of simulated magnetization images generated through the atomistic Monte Carlo framework described in Section 3.2. While the simulation protocol incorporates physically realistic finite-size effects, cylindrical boundary conditions, and thermodynamic relaxation via simulated annealing, it does not account for several experimental phenomena that are routinely present in real magnetic nanodot imaging. These include thermal noise and instrument-specific point spread functions inherent to the imaging apparatus, crystallographic defects, vacancies, and grain boundaries that break the assumed periodic lattice symmetry, surface reconstruction effects that modify the effective exchange and DMI parameters at the physical boundary of the nanodot, and substrate-induced strain fields that can renormalize the bulk Hamiltonian parameters. As a consequence, a model trained exclusively on simulated data may exhibit significant performance degradation when applied to experimentally acquired magnetization images — a domain shift problem that has not been quantified in the present work. Bridging this simulation-to-experiment gap would require either transfer learning with a small set of experimentally labeled samples or the incorporation of physics-informed data augmentation strategies that realistically replicate experimental noise sources.

Although the dataset encompasses 218,256 simulated configurations spanning a broad range of Hamiltonian parameters, the sampling strategy employed — uniform random sampling within the parameter bounds — does not guarantee uniform coverage of the magnetic phase space. As evidenced by the UMAP clustering analysis of Section 5.2, the latent space exhibits strongly non-uniform density: the Labyrinthine & Conical and Helical phases, which require precise combinations of $\tilde{K}_{DM}$ and $T^{(0)}$, are substantially overrepresented relative to the Ferromagnetic phase, which occupies a disproportionately small region of the parameter space. This imbalance propagates directly into the regression performance: the models achieve higher accuracy for phases with greater dataset density, while underperforming in topologically marginal regimes such as the ferromagnetic-to-skyrmion phase boundary where the mapping from parameters to magnetization textures is most degenerate.

From a deep learning perspective, the RAM-based interpretability framework introduced in this work provides spatially resolved gradient attributions that are physically consistent and architecturally agnostic. However, RAMs share the fundamental limitations of all gradient-based saliency methods: the attribution maps reflect the local sensitivity of the objective function O_p to spatial activations at the point of evaluation, but do not provide a globally faithful explanation of the model's decision function across the full parameter space. In particular, gradient saturation in deeply nonlinear networks can produce spatially diffuse or spurious attributions for parameters with low predictive signal — precisely the degenerate parameters identified in this study. Furthermore, the current RAM formulation computes a single attribution map per image per parameter, which does not capture the stochastic uncertainty in the attribution due to the non-deterministic elements of the forward pass. Future extensions could incorporate ensemble-based or Bayesian attribution methods to provide confidence intervals on the spatial interpretability maps, which would be particularly valuable for the physically ambiguous parameter–phase combinations identified in Section 5.3.

While the inference latency of the regression models is highly competitive — sub-2 ms per image at batch size $B = 128$ for both the Vision Transformer and Xception — the computation of Regression Activation Maps introduces a substantial additional overhead. Each RAM requires a forward pass, a backward pass through the gradient tape, and a spatial upsampling operation, increasing the per-image computational cost by approximately one to two orders of magnitude relative to standard inference. For the Vision Transformer, which requires constructing a tf.Variable over the $N_r = 196$ token embeddings and differentiating through the GlobalAveragePooling and Dense layers, this overhead is particularly pronounced. Consequently, the full RAMs pipeline is better suited to offline post-hoc analysis of selected samples rather than real-time interpretability during high-throughput characterization. Approximation strategies such as layer-wise relevance propagation or integrated gradients computed on compressed token representations could reduce this overhead while preserving the physical interpretability of the attribution maps.

6. Conclusions

This work introduced a regression-based explainable deep learning framework for the inverse estimation of Hamiltonian parameters from two-dimensional magnetization images of magnetic nanodots. The proposed methodology integrates atomistic Monte Carlo simulations under a physically rigorous extended Heisenberg Hamiltonian with eight free parameters, five state-of-the-art deep learning architectures, an unsupervised latent space analysis via UMAP and density-based clustering, and a novel spatial interpretability formulation RAMs that generalizes gradient-based activation mapping to the continuous regression domain for both convolutional and transformer-based architectures.

Among the five evaluated architectures, the Vision Transformer achieved the highest overall regression accuracy across all eight Hamiltonian parameters, followed by Xception as the best-performing CNN baseline. Both models demonstrated strong predictive capability for the thermodynamic temperature $T^{(0)}$, the second-shell exchange constant $\tilde{J}_2$, and the DMI strength $\tilde{K}_{DM}$, which constitute the physically dominant parameters governing the observable spin texture in the s_z projection. By contrast, the magnetocrystalline anisotropy constants $\tilde{K}_{an1}$ and $\tilde{K}_{anS}$, the external Zeeman field $\tilde{H}_{ex}$, and the higher-order exchange constants $\tilde{J}_3$ and $\tilde{J}_4$ exhibited systematically low identifiability across all architectures, with near-zero or undefined R^2 values. This result establishes that the fundamental limitation of the inverse problem is not architectural capacity but the intrinsic partial observability of the extended Hamiltonian from the two-dimensional s_z projection — a physical constraint that cannot be overcome by increasing model complexity.

Beyond the aggregate regression performance, the UMAP projection of the Vision Transformer's latent representations revealed that the model has autonomously organized the magnetization textures into geometrically distinct and physically interpretable regions of the embedding space, without any explicit supervision of magnetic phase labels. Density-based clustering of this embedding identified four dominant magnetic phase categories: Ferromagnetic, Skyrmions and Others, Labyrinthine & Conical, and Helical. The per-phase regression analysis confirmed that the model achieves its highest accuracy within the ordered low-temperature phases — Labyrinthine & Conical ($R^2 = 0.978$ for $T^{(0)}$, $R^2 = 0.970$ for $\tilde{K}_{DM}$) and Helical ($R^2 = 0.968$ for $T^{(0)}$, $R^2 = 0.866$ for $\tilde{K}_{DM}$) — where the magnetization texture carries the most discriminative physical information. These results demonstrate that the Vision Transformer's global self-attention mechanism effectively encodes the long-range spin correlations characteristic of topologically ordered magnetic phases.

The proposed RAMs formulation successfully extended gradient-based activation mapping to the continuous regression setting through a scale-invariant Gaussian error-decay objective $O_p = \exp(-\gamma_p(\theta_p - \hat{\theta}_p)^2)$, providing physically interpretable spatial attribution maps for each Hamiltonian parameter independently. For the Xception architecture, the layer-resolved RAM analysis revealed a physically consistent hierarchical decoupling across network depth: early entry-flow layers (block1, block3) capture fine-grained domain periodicity and chiral winding angles at high spatial resolution, while deeper middle-flow and exit-flow layers (block8–block13) consolidate these local features into coarser, topologically meaningful representations of the global magnetic phase. For the Vision Transformer, the patch-level RAM attributions derived from the reshaped token embedding $\mathbf{A}_{\text{ViT}} \in \mathbb{R}^{14\times14\times768}$ exhibited smooth, globally coherent spatial patterns that concentrated on chiral domain walls and stripe boundaries in the ordered phases, and produced sparse, peripheral activations for physically degenerate parameter–phase combinations. Critically, both architectures identified the same set of physically meaningful spatial features as the primary drivers of accurate regression, confirming that the learned representations are physically grounded rather than statistically arbitrary, and both failed to produce spatially concentrated activations for the non-identifiable parameters, providing a direct, architecture-agnostic validation of the identifiability hierarchy.

From a deployment standpoint, the inference latency analysis demonstrated that the proposed framework is computationally viable for practical materials characterization workflows. Xception achieves the lowest per-image latency in both the single-sample (127.71 ms) and batch (1.47 ms at $B = 128$) regimes, while the Vision Transformer achieves competitive batch throughput at 1.92 ms/image at $B = 128$ — a factor of only 1.31× slower than Xception despite its 4.1× larger parameter count. These results establish that the Vision Transformer is the preferred architecture for high-throughput batch characterization pipelines, where its superior regression accuracy is achieved at a negligible throughput cost relative to the best CNN baseline.

Taken together, the results of this study demonstrate that regression-based deep learning, combined with physically motivated interpretability methods, constitutes a viable and physically meaningful approach to the inverse Hamiltonian estimation problem in magnetic nanostructures. The proposed RAMs framework is architecture-agnostic and directly applicable to any continuous regression task where spatial attribution of physical parameters is required, extending beyond magnetic nanodots to other physical systems characterized by spatially resolved imaging modalities.

Several directions merit investigation in future work. First, incorporating the full three-dimensional spin vector field $\mathbf{S}_i = (s_x, s_y, s_z)$ or multi-layer volumetric magnetization images as input would substantially reduce the partial observability constraint and improve the identifiability of the anisotropy and field parameters. Second, domain adaptation strategies, including physics-informed data augmentation and few-shot transfer learning from experimentally labeled samples, are necessary to bridge the simulation-to-experiment gap and enable direct application to real magnetic imaging data. Third, extending the RAM formulation to incorporate uncertainty quantification through ensemble-based or Bayesian attribution methods would provide confidence intervals on the spatial interpretability maps, which are particularly valuable for physically ambiguous parameter–phase combinations near topological phase boundaries. Finally, the integration of the proposed framework with active learning strategies, where the model identifies the most informative parameter combinations to simulate next, could dramatically accelerate the construction of training datasets for high-dimensional Hamiltonian spaces, enabling the extension of the inverse estimation approach to more complex magnetic systems with larger numbers of competing energetic contributions.

Acknowledgments

Under grants provided by the research project: "Aprendizaje automático aplicado a la predicción eficiente de propiedades magnéticas de nanopartículas como soporte a la generación de dispositivos electrónicos" Hermes 62642, funded by Universidad Nacional de Colombia. Also, A. Alvarez thanks to the project: "Estimación de la capacidad de generación de energía eléctrica del agua coproducida en campos de petróleo y gas en Colombia a partir de técnicas de aprendizaje automático informado por la física," Code 111908, funded by 951-2024 Convocatoria fortalecimiento del conocimiento científico y tecnológico de las fuentes - Minciencias.ñkm

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Gemini and Claude in order to enhance the writing style. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

References

[1] A. J. Heinrich, W. D. Oliver, L. M. Vandersypen, A. Ardavan, R. Sessoli, D. Loss, A. B. Jayich, J. Fernandez-Rossier, A. Laucht, and A. Morello, "Quantum-coherent nanoscience," Nature nanotechnology, vol. 16, no. 12, pp. 1318–1329, 2021.

[2] K. Wang, V. Bheemarasetty, J. Duan, S. Zhou, and G. Xiao, "Fundamental physics and applications of skyrmions: A review," Journal of Magnetism and Magnetic Materials, vol. 563, p. 169905, 2022.

[3] A. Fert, R. Ramesh, V. Garcia, F. Casanova, and M. Bibes, "Electrical control of magnetism by electric field and current-induced torques," Reviews of Modern Physics, vol. 96, no. 1, p. 015005, 2024.

[4] B. Issa, I. M. Obaidat, B. A. Albiss, and Y. Haik, "Magnetic nanoparticles: surface effects and properties related to biomedicine applications," International journal of molecular sciences, vol. 14, no. 11, pp. 21266–21305, 2013.

[5] I. Charalampidis and J. Barker, "Metadynamics calculations of the effect of thermal spin fluctuations on skyrmion stability," arXiv preprint arXiv:2310.03169, 2023.

[6] X. Yua, M. Mostovoyb, Y. Tokunagaa, W. Zhangc, K. Kimotoc, Y. Matsuic, Y. Kanekod, N. Nagaosaa, and Y. Tokuraa, "Magnetic stripes and skyrmions with helicity reversals," PNAS, vol. 114, no. 2, p. E265, 2017.

[7] M. Birch, D. Cortés-Ortuño, L. Turnbull, M. Wilson, F. Groß, N. Träger, A. Laurenson, N. Bukin, S. Moody, M. Weigand, et al., "Real-space imaging of confined magnetic skyrmion tubes," Nature communications, vol. 11, no. 1, p. 1726, 2020.

[8] X. Yu, "Magnetic imaging of various topological spin textures and their dynamics," Journal of Magnetism and Magnetic Materials, vol. 539, p. 168332, 2021.

[9] H. Zhang, Y. Zhang, Z. Hou, M. Qin, X. Gao, and J. Liu, "Magnetic skyrmions: materials, manipulation, detection, and applications in spintronic devices," Materials Futures, vol. 2, no. 3, p. 032201, 2023.

[10] Y. Tokura and N. Kanazawa, "Magnetic skyrmion materials," Chemical Reviews, vol. 121, no. 5, pp. 2857–2897, 2020.

[11] M. Zelent, M. Moalic, M. Mruczkiewicz, X. Li, Y. Zhou, and M. Krawczyk, "Stabilization and racetrack application of asymmetric néel skyrmions in hybrid nanostructures," Scientific Reports, vol. 13, no. 1, p. 13572, 2023.

[12] S. Rohart and A. Thiaville, "Skyrmion confinement in ultrathin film nanostructures in the presence of dzyaloshinskii-moriya interaction," Physical Review B—Condensed Matter and Materials Physics, vol. 88, no. 18, p. 184422, 2013.

[13] J. F. Kong, Y. Ren, M. N. Tey, P. Ho, K. H. Khoo, X. Chen, and A. Soumyanarayanan, "Quantifying the magnetic interactions governing chiral spin textures using deep neural networks," ACS Applied Materials & Interfaces, vol. 16, no. 1, pp. 1025–1032, 2023.

[14] R. F. Evans, W. J. Fan, P. Chureemart, T. A. Ostler, M. O. Ellis, and R. W. Chantrell, "Atomistic spin model simulations of magnetic nanomaterials," Journal of Physics: Condensed Matter, vol. 26, no. 10, p. 103202, 2014.

[15] A. Aldarawsheh, M. Sallermann, M. Abusaa, and S. Lounis, "Intrinsic néel antiferromagnetic multimeronic spin textures in ultrathin films," The journal of physical chemistry letters, vol. 14, no. 40, pp. 8970–8978, 2023.

[16] G. Finocchio, F. Büttner, R. Tomasello, M. Carpentieri, and M. Kläui, "Magnetic skyrmions: from fundamental to applications," Journal of Physics D: Applied Physics, vol. 49, no. 42, p. 423001, 2016.

[17] C. Back, V. Cros, H. Ebert, K. Everschor-Sitte, A. Fert, M. Garst, T. Ma, S. Mankovsky, T. Monchesky, M. Mostovoy, et al., "The 2020 skyrmionics roadmap," Journal of Physics D: Applied Physics, vol. 53, no. 36, p. 363001, 2020.

[18] Z. He, W. Du, K. Dou, Y. Dai, B. Huang, and Y. Ma, "Ferroelectrically tunable magnetic skyrmions in two-dimensional multiferroics," Materials Horizons, vol. 10, no. 9, pp. 3450–3457, 2023.

[19] R. Murakami, M. Mizumaki, I. Akai, and H. Shouno, "Inverse estimation of parameters for the magnetic domain via dynamics matching using visual-perceptive similarity," Science and Technology of Advanced Materials: Methods, vol. 2, no. 1, pp. 139–152, 2022.

[20] H. Y. Kwon, H. Yoon, C. Lee, G. Chen, K. Liu, A. Schmid, Y. Wu, J. Choi, and C. Won, "Magnetic hamiltonian parameter estimation using deep learning techniques," Science advances, vol. 6, no. 39, p. eabb0872, 2020.

[21] T. N. H. Nguyen, L. Rasabathina, O. Hellwig, A. Sharma, G. Salvan, S. Yochelis, Y. Paltiel, L. T. Baczewski, and C. Tegenkamp, "Cooperative effect of electron spin polarization in chiral molecules studied with non-spin-polarized scanning tunneling microscopy," ACS applied materials & interfaces, vol. 14, no. 33, pp. 38013–38020, 2022.

[22] J. Zhao, L. Bai, S. Li, Z. Cao, Y. Peng, J. Bai, X. Cai, X. Shi, X. Lin, G. Wei, et al., "An overview of advanced instruments for magnetic characterization and measurements," Frontiers in Electronics, vol. 6, p. 1645594, 2025.

[23] J. H. Gaida, H. Lourenço-Martins, S. V. Yalunin, A. Feist, M. Sivis, T. Hohage, F. J. García de Abajo, and C. Ropers, "Lorentz microscopy of optical fields," Nature Communications, vol. 14, no. 1, p. 6545, 2023.

[24] M. d'Aquino, S. Perna, M. Pancaldi, R. Hertel, S. Bonetti, and C. Serpico, "Micromagnetic study of inertial spin waves in ferromagnetic nanodots," Physical Review B, vol. 107, no. 14, p. 144412, 2023.

[25] S. Lepadatu, "Micromagnetic monte carlo method with variable magnetization length based on the landau–lifshitz–bloch equation for computation of large-scale thermodynamic equilibrium states," Journal of Applied Physics, vol. 130, no. 16, 2021.

[26] X. Yin, Computational Materials Design: Integrating Physics With Optimization and Machine Learning. PhD thesis, Carnegie Mellon University, 2023.

[27] S. Karnatov, "Analysis of psnr, ssim, lpips metrics in the context of human perception of visual similarity," Transport systems and technologies, no. 46, 2025.

[28] W. S. Lee, T. Song, and K.-M. Kim, "Deep learning methods for hamiltonian parameter estimation and magnetic domain image generation in twisted van der waals magnets," Machine Learning: Science and Technology, vol. 5, no. 2, p. 025073, 2024.

[29] Y. Chen, Q. Yang, Y. Li, H. Zhang, and C. Zhang, "Data-driven deep convolutional neural networks for electromagnetic field estimation of transformers," IEEE Transactions on Applied Superconductivity, vol. 34, no. 8, pp. 1–5, 2024.

[30] S. Lan, S. Li, and B. Shahbaba, "Scaling up bayesian uncertainty quantification for inverse problems using deep neural networks," SIAM/ASA Journal on Uncertainty Quantification, vol. 10, no. 4, pp. 1684–1713, 2022.

[31] L. Huang, J. Li, X. Ding, Y. Zhang, H. Chen, and A. Ozcan, "Cycle-consistency-based uncertainty quantification of neural networks in inverse imaging problems," Intelligent Computing, vol. 2, p. 0071, 2023.

[32] M. Baldan, P. Di Barba, and D. A. Lowther, "Physics-informed neural networks for inverse electromagnetic problems," IEEE Transactions on Magnetics, vol. 59, no. 5, pp. 1–5, 2023.

[33] W. Wang, Z. Wang, Y. Zhang, B. Sun, and K. Xia, "Learning order parameters from videos of skyrmion dynamical phases with neural networks," Physical Review Applied, vol. 16, no. 1, p. 014005, 2021.

[34] T. Čadež and K.-M. Kim, "Neural scaling laws for deep regression on domain image data of twisted magnets," Machine Learning: Science and Technology, vol. 7, no. 2, p. 025011, 2026.

[35] M. He, B. Li, and S. Sun, "A survey of class activation mapping for the interpretability of convolution neural networks," in International Conference On Signal And Information Processing, Networking And Computers, pp. 399–407, Springer, 2022.

[36] J. Zeng, M. Albooyeh, M. Rajaei, A. A. Sifat, E. O. Potma, H. K. Wickramasinghe, and F. Capolino, "Direct detection of photoinduced magnetic force at the nanoscale reveals magnetic nearfield of structured light," Science advances, vol. 8, no. 45,

p. eadd0233, 2022.
[37] A. Altaf, M. Shakir, H. A. Irshad, S. Atif, U. Kumari, O. Islam, W. T. Kimberly, E. Knopp, C. Truwit, K. Siddiqui, et al., "Applications, limitations and advancements of ultra-low-field magnetic resonance imaging: a scoping review," Surgical neurology international, vol. 15, p. 218, 2024.
[38] G. A. Silva, A. Bergeron, J. Plascak, and D. Landau, "Three-dimensional heisenberg model with dzyaloshinskii-moriya interaction: A monte carlo study," Physical Review E, vol. 109, no. 6, p. 064113, 2024.
[39] W. T. Mongwe, R. Mbuvha, and T. Marwala, "Adaptive magnetic hamiltonian monte carlo," IEEE Access, vol. 9, pp. 152993–153003, 2021.
[40] A. Talapatra, U. Gajera, S. P. P, J. Arout Chelvane, and J. R. Mohanty, "Understanding the magnetic microstructure through experiments and machine learning algorithms," ACS applied materials & interfaces, vol. 14, no. 44, pp. 50318–50330, 2022.
[41] M. Liu, D. Grana, K. Mosegaard, M. K. Sen, M. Xu, and T. Mukerji, "Bayesian inference for subsurface geophysical inverse problems," Reviews of Geophysics, vol. 64, no. 1, p. e2025RG000884, 2026.
[42] Z. Zhao, J. C. Ye, and Y. Bresler, "Generative models for inverse imaging problems: From mathematical foundations to physics-driven applications," IEEE Signal Processing Magazine, vol. 40, no. 1, pp. 148–163, 2023.
[43] A. Shmakov, K. Greif, M. Fenton, A. Ghosh, P. Baldi, and D. Whiteson, "End-to-end latent variational diffusion models for inverse problems in high energy physics," Advances in neural information processing systems, vol. 36, pp. 65102–65127, 2023.
[44] C. Sui, Z. Xu, Y. Meng, H. Wang, B. Zhang, and G. Vivone, "Cyclemamba: Cycle-consistent learning for aerial visible-to-infrared image translation," IEEE Transactions on Geoscience and Remote Sensing, 2026.
[45] A. Will-Cole, A. G. Kusne, P. Tonner, C. Dong, X. Liang, H. Chen, and N. X. Sun, "Application of bayesian optimization and regression analysis to ferromagnetic materials development," IEEE transactions on magnetics, vol. 58, no. 1, pp. 1–8, 2021.
[46] B. Sridharan, M. Goel, and U. D. Priyakumar, "Modern machine learning for tackling inverse problems in chemistry: molecular design to realization," Chemical communications, vol. 58, no. 35, pp. 5316–5331, 2022.
[47] J. Agudelo-Giraldo, E. Restrepo-Parra, and D. Landau, "A study of magnetic domains in thin films from fm and afm interactions," Physica A: Statistical Mechanics and its Applications, vol. 517, pp. 542–550, 2019.
[48] V. Borisov, Y. O. Kvashnin, N. Ntallis, D. Thonig, P. Thunström, M. Pereiro, A. Bergman, E. Sjöqvist, A. Delin, L. Nordström, et al., "Heisenberg and anisotropic exchange interactions in magnetic materials with correlated electronic structure and significant spin-orbit coupling," Physical Review B, vol. 103, no. 17, p. 174422, 2021.
[49] T. Moriya, "Anisotropic superexchange interaction and weak ferromagnetism," Physical review, vol. 120, no. 1, p. 91, 1960.
[50] O. Eriksson, A. Bergman, L. Bergqvist, and J. Hellsvik, Atomistic spin dynamics: foundations and applications. Oxford university press, 2017.
[51] B. D. Cullity and C. D. Graham, Introduction to magnetic materials. John Wiley & Sons, 2011.
[52] J. M. Coey, Magnetism and magnetic materials. Cambridge university press, 2010.
[53] A. Fert, N. Reyren, and V. Cros, "Magnetic skyrmions: advances in physics and potential applications," Nature Reviews Materials, vol. 2, no. 7, p. 17031, 2017.
[54] D. Landau and K. Binder, A guide to Monte Carlo simulations in statistical physics. Cambridge university press, 2021.
[55] L. Desplat, D. Suess, J.-V. Kim, and R. L. Stamps, "Thermal stability of metastable magnetic skyrmions: Entropic narrowing and significance of internal eigenmodes," Physical Review B, vol. 98, no. 13, p. 134407, 2018.
[56] K. P. Murphy, Probabilistic machine learning: Advanced topics. MIT press, 2023.
[57] B. Koonce, "Resnet 50," in Convolutional neural networks with swift for tensorflow: image recognition and dataset categorization, pp. 63–72, Springer, 2021.
[58] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in Proceedings of the IEEE conference on computer vision and pattern recognition, pp. 4700–4708, 2017.
[59] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in Proceedings of the IEEE conference on computer vision and pattern recognition, pp. 1251–1258, 2017.
[60] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al., "An image is worth 16x16 words: Transformers for image recognition at scale," arXiv preprint arXiv:2010.11929, 2020.
[61] K. Vinogradova, A. Dibrov, and G. Myers, "Towards interpretable semantic segmentation via gradient-weighted class activation mapping (student abstract)," in Proceedings of the AAAI Conference on Artificial Intelligence, 2020.
[62] Y. Zhang, Y. Zhu, J. Liu, and W. Yu, "An interpretability optimization method for deep learning networks based on gradcam," IEEE, 2024.
[63] N. Makke and S. Chawla, "Interpretable scientific discovery with symbolic regression: a review," Applied Intelligence Review, 2024.
[64] I. Radosavovic, T. Xiao, S. James, P. Abbeel, J. Malik, and T. Darrell, "Real-world robot learning with masked visual pre-training," in Conference on Robot Learning, pp. 416–426, PMLR, 2023.
[65] B. Ghojogh, A. Ghodsi, F. Karray, and M. Crowley, "Uniform manifold approximation and projection (umap) and its variants: tutorial and survey," arXiv preprint arXiv:2109.02508, 2021.